Principles of Quantum Mechanics

Part II

Recorded from the lecturer in 2026-2027.

0. Introduction

∗ Recall features from IB QM (Supplementary Notes 1)

- Wave-particle duality:

  Waves behaving like particles, e.g. light quanta, & vice versa (particles behaving like waves).

- Wavefunction $\psi(x)$ for a particle — prob. density $|\psi(x)|^2$.

  (Prob. intrinsic to theory.)

- Observables became operators on wavefns:

$$\hat{x} \to x, \qquad \hat{p} \to -i\hbar \frac{\partial}{\partial x}.$$

  & lack of commutativity, e.g.

$$[\hat{x}, \hat{p}] = i\hbar$$

  leads to Uncertainty Relation.

- Schrödinger eqn with Hamiltonian H specifies dynamics & eigenvalues of H are allowed energy levels of system.

- All of this explains H atom structure & resulting line spectra.

∗ Aims of this course:

- Reformulate QM in a more powerful, flexible, & useful form — the Dirac formalism (go beyond wavefns!).

Previous description appears as particular realisation & also leads to alternatives, e.g.

$$\tilde{\psi}(p) \qquad \text{with} \qquad \hat{x} \to +i\hbar \frac{\partial}{\partial p} \quad \& \quad \hat{p} \to p.$$

This more general approach allows

- Simpler analysis of known problems, e.g. oscillator

but also lots of new things

- Spin of particles
- Symmetries (translations, rotations, for example) & conservation laws.
- Identical particles (bosons & fermions)
- Framework for quantising more general systems, e.g. EM field
- Perturbation theory: practical tool for computations — we will not look at too many detailed applications, but we will try to keep track of what all this is for — physical predictions

1. Dirac Formalism

1.1 Concepts & Definitions

A quantum system is described at each instant by a state $|\psi\rangle$ which belongs to a complex vector space V; any non-zero element of this space V is a possible state.

Thus

$$|\psi\rangle \;\&\; |\phi\rangle \in V \quad \Rightarrow \quad \alpha|\psi\rangle + \beta|\phi\rangle \in V \qquad \text{for any } \alpha, \beta \in \mathbb{C}.$$

Physically this is a superposition principle leading to wave-like behaviour, but these states are not wavefns. There are also dual or conjugate states $\langle\phi|$ which belong to the dual space $V^\dagger$.

By definition, states & duals can be combined or paired to give a complex number

$$\underset{\text{"bra"}}{\langle\phi|}\,, \quad \underset{\text{"ket"}}{|\psi\rangle} \longmapsto \underset{\text{"bra(c)ket"}}{\langle\phi|\psi\rangle}\,, \qquad V^\dagger \times V \to \mathbb{C},$$

with

$$\langle\phi|\big(\alpha_1|\psi_1\rangle + \alpha_2|\psi_2\rangle\big) = \alpha_1\langle\phi|\psi_1\rangle + \alpha_2\langle\phi|\psi_2\rangle,$$
$$\big(\beta_1\langle\phi_1| + \beta_2\langle\phi_2|\big)|\psi\rangle = \beta_1\langle\phi_1|\psi\rangle + \beta_2\langle\phi_2|\psi\rangle.$$

The spaces V & $V^\dagger$ come with an inner product, which can be described as a one-to-one correspondence between the spaces

$$V \longleftrightarrow V^\dagger,$$

with

$$|\psi\rangle \longleftrightarrow \langle\psi| = \big(|\psi\rangle\big)^\dagger \qquad \text{(use same symbol for corresponding states).}$$

The inner product is

$$|\phi\rangle,\ |\psi\rangle \longmapsto \langle\phi|\psi\rangle = \big(|\phi\rangle\big)^\dagger|\psi\rangle, \qquad V \times V \to \mathbb{C},$$

& it is assumed to obey

$$\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^* \qquad \text{hermitian},$$
$$\big\|\,|\psi\rangle\,\big\|^2 = \langle\psi|\psi\rangle \qquad \text{real},$$
$$\left.\begin{array}{l}\langle\psi|\psi\rangle \geq 0,\\ \langle\psi|\psi\rangle = 0 \quad \text{iff} \quad |\psi\rangle = 0\end{array}\right\} \qquad \text{positive definite},$$

where now $\big\|\,|\psi\rangle\,\big\|$ is called the norm.

Note: Knowing $\langle\phi|\psi\rangle$ for all $\langle\phi|$ determines $|\psi\rangle$ (& vice versa).

[Check:

$$\langle\phi|\psi_1\rangle = \langle\phi|\psi_2\rangle \quad \Rightarrow \quad \langle\phi|\chi\rangle = 0, \qquad \text{where } |\chi\rangle = |\psi_1\rangle - |\psi_2\rangle.$$

If this holds $\forall\,\langle\phi|$, set $\langle\phi| = \langle\chi|$ & get

$$\langle\chi|\chi\rangle = \big\|\,|\chi\rangle\,\big\|^2 = 0 \quad \Rightarrow \quad |\chi\rangle = 0, \qquad \text{as claimed.}$$

]

The physical content of any state is unchanged by

$$|\psi\rangle \to \alpha|\psi\rangle \qquad \text{for any } \alpha \in \mathbb{C}, \quad \alpha \neq 0.$$

We will usually normalise to make $\big\|\,|\psi\rangle\,\big\| = 1$, but can still change

$$|\psi\rangle \to e^{i\theta}|\psi\rangle.$$

Relative phases can be important in expressions like $\alpha|\psi\rangle + \beta|\phi\rangle$.

The space V is complete: we assume that appropriate sequences & series converge. A complete inner product space is called a Hilbert space.

An operator Q is a linear map on states

$$|\psi\rangle \longmapsto Q|\psi\rangle, \qquad V \to V,$$

with

$$Q\big(\alpha|\psi\rangle + \beta|\phi\rangle\big) = \alpha Q|\psi\rangle + \beta Q|\phi\rangle.$$

The same operator can be thought of as acting to the left on dual states,

$$\langle\phi| \longmapsto \langle\phi|Q, \qquad V^\dagger \to V^\dagger,$$

where the right-hand side is defined by

$$\big(\langle\phi|Q\big)|\psi\rangle = \langle\phi|\big(Q|\psi\rangle\big) = \langle\phi|Q|\psi\rangle.$$

End of lecture 1

Last time: abstract notation for states

$$|\psi\rangle \in V, \qquad \langle\phi| \in V^\dagger, \qquad \& \text{ map } V \longleftrightarrow V^\dagger,$$

& operators acting, e.g. $|\psi\rangle \longmapsto Q|\psi\rangle$ (§ 1.1).

Note: States & operators $|\psi\rangle$, $\langle\phi|$, Q can be dealt with by rules that mirror those for complex vectors $\underline{u}$ (column), $\underline{v}^\dagger$ (row), & a matrix A; e.g. compare

$$\langle\phi|Q|\psi\rangle \qquad \text{with} \qquad \underline{v}^\dagger A \underline{u}.$$

For any operator Q, say $|\psi\rangle \neq 0$ is an eigenstate with eigenvalue λ if

$$Q|\psi\rangle = \lambda|\psi\rangle;$$

equivalently,

$$\langle\psi|Q^\dagger = \lambda^*\langle\psi|.$$

For general Q, λ may be complex.

For any Q, the hermitian conjugate or adjoint $Q^\dagger$ is defined by

$$\langle\phi|Q^\dagger = \big(Q|\phi\rangle\big)^\dagger \qquad \text{for any } |\phi\rangle.$$

Equivalently,

$$\begin{aligned}\langle\phi|Q^\dagger|\psi\rangle &= \big(Q|\phi\rangle\big)^\dagger|\psi\rangle \\ &= \langle\psi|Q|\phi\rangle^* \qquad \text{for all } |\phi\rangle,\ |\psi\rangle.\end{aligned}$$

Simple consequences:

$$\begin{aligned}(\alpha A + \beta B)^\dagger &= \alpha^* A^\dagger + \beta^* B^\dagger, \\ (AB)^\dagger &= B^\dagger A^\dagger,\end{aligned}$$

for any operators A, B & $\alpha, \beta \in \mathbb{C}$, &

$$\big(Q^\dagger\big)^\dagger = Q.$$

Observables

An operator Q is hermitian or self-adjoint if

$$Q^\dagger = Q;$$

such operators are called observables because they correspond to physical quantities that can be measured, e.g. position, momentum, energy.

To define a quantum system, we must specify the Hamiltonian H whose eigenvalues are the allowed energy levels of the system:

$$H|\chi\rangle = E|\chi\rangle \qquad \text{energy eigenvalue problem.}$$

Key results for any hermitian Q:

(i) All eigenvalues are real.

(ii) Eigenstates with distinct eigenvalues are orthogonal.

(iii) Any state can be expanded in terms of (written as a linear combination of) eigenstates.

We prove (i) & (ii) & assume (iii).

(i) Suppose

$$Q|\psi\rangle = \lambda|\psi\rangle.$$

Then

$$\langle\psi|Q^\dagger = \lambda^*\langle\psi| = \langle\psi|Q \qquad (Q^\dagger = Q).$$

Hence

$$\langle\psi|Q|\psi\rangle = \lambda\langle\psi|\psi\rangle = \lambda^*\langle\psi|\psi\rangle.$$

But

$$\big\| \, |\psi\rangle \, \big\|^2 = \langle\psi|\psi\rangle \neq 0 \qquad (|\psi\rangle \neq 0),$$

so $\lambda^* = \lambda$.

(ii) Let $|n\rangle$ be eigenstates of Q with eigenvalues $\lambda = q_n$ (real), where n is some discrete label, possibly ranging over infinitely many values.

$$Q|n\rangle = q_n|n\rangle, \qquad Q|m\rangle = q_m|m\rangle,$$

or

$$\langle m|Q = q_m\langle m| \qquad (q_m^* = q_m).$$

Hence

$$\langle m|Q|n\rangle = q_n\langle m|n\rangle = q_m\langle m|n\rangle.$$

But $q_m \neq q_n$, so $\langle m|n\rangle = 0$.

1.2 The Harmonic Oscillator

A one-dimensional harmonic oscillator of mass m & frequency ω is a system with position $\hat{x}$, momentum $\hat{p}$ & Hamiltonian

$$H = \frac{1}{2m}\hat{p}^{\,2} + \frac{1}{2}m\omega^2\hat{x}^{\,2}.$$

Seek eigenstates & eigenvalues of H.

(a) Annihilation, creation & number operators

Define

$$\begin{cases} a = \left(\frac{m\omega}{2\hbar}\right)^{1/2}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right), \\ a^\dagger = \left(\frac{m\omega}{2\hbar}\right)^{1/2}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right). \end{cases}$$

These are dimensionless hermitian conjugates, since $\hat{x}^\dagger = \hat{x}$, $\hat{p}^\dagger = \hat{p}$. Equivalently,

$$\begin{cases} \hat{x} = \left(\frac{\hbar}{2m\omega}\right)^{1/2}(a + a^\dagger), \\ \hat{p} = \left(\frac{\hbar m\omega}{2}\right)^{1/2} i(a^\dagger - a). \end{cases}$$

Easy to check:

$$[\hat{x}, \hat{p}] = i\hbar \quad \Longleftrightarrow \quad [a, a^\dagger] = 1.$$

In fact,

$$\begin{aligned} aa^\dagger &= \frac{m\omega}{2\hbar}\hat{x}^{\,2} + \frac{1}{2m\omega\hbar}\hat{p}^{\,2} - \frac{i}{2\hbar}(\hat{x}\hat{p} - \hat{p}\hat{x}) \\ &= \frac{1}{\hbar\omega}H + \frac{1}{2}, \end{aligned}$$

but

$$a^\dagger a = \frac{1}{\hbar\omega}H - \frac{1}{2}.$$

This confirms the commutation relation, but we also find

$$H = \hbar\omega\left(N + \frac{1}{2}\right),$$

with $N = a^\dagger a$ the number operator.

$N^\dagger = N$, & finding eigenvalues & eigenstates of N is equivalent to finding those for H. Let $|\lambda\rangle$ be any eigenstate of N with eigenvalue λ:

$$N|\lambda\rangle = \lambda|\lambda\rangle \quad \Longleftrightarrow \quad H|\lambda\rangle = E|\lambda\rangle, \qquad E = \hbar\omega\left(\lambda + \frac{1}{2}\right).$$

Then, assuming $\big\|\,|\lambda\rangle\,\big\|^2 = 1$ (normalised),

$$\begin{aligned} \lambda = \langle\lambda|N|\lambda\rangle &= \langle\lambda|a^\dagger a|\lambda\rangle = \big(a|\lambda\rangle\big)^\dagger\big(a|\lambda\rangle\big) \\ &= \big\|a|\lambda\rangle\big\|^2. \end{aligned}$$

Thus $\lambda \geq 0$ (all eigenvalues are non-negative), & $\lambda = 0$ iff $a|\lambda\rangle = 0$.

Now consider

$$\begin{aligned} N\big(a^\dagger|\lambda\rangle\big) &= \big([N, a^\dagger] + a^\dagger N\big)|\lambda\rangle \\ &= \big(a^\dagger + a^\dagger\lambda\big)|\lambda\rangle \\ &= (\lambda + 1)\big(a^\dagger|\lambda\rangle\big). \end{aligned}$$

Similarly,

$$\begin{aligned} N\big(a|\lambda\rangle\big) &= \big([N, a] + aN\big)|\lambda\rangle \\ &= (-a + a\lambda)|\lambda\rangle \\ &= (\lambda - 1)a|\lambda\rangle. \end{aligned}$$

Here we have used the results for commutators

$$[N, a^\dagger] = a^\dagger \qquad \big(= [a^\dagger a, a^\dagger] = a^\dagger [a, a^\dagger]\big),$$

&

$$[N, a] = -a.$$

New states will be eigenstates provided their norms are non-zero. Compute:

$$\big\| a|\lambda\rangle \big\|^2 = \lambda,$$

$$\begin{aligned} \big\| a^\dagger|\lambda\rangle \big\|^2 &= \langle\lambda| a a^\dagger |\lambda\rangle = \langle\lambda| a^\dagger a + 1 |\lambda\rangle \\ &= \lambda + 1. \end{aligned}$$

Therefore $a^\dagger$ & a are called <u>creation</u> (or raising) & <u>annihilation</u> (or lowering) operators, respectively. Since all eigenvalues are non-negative, any non-integer λ would eventually lead to a negative eigenvalue under repeated lowering, which is impossible. Thus, λ must be a non-negative integer. Repeated lowering then reaches the ground state $|0\rangle$, where $a|0\rangle = 0$. Since the raising operator $a^\dagger$ can act indefinitely, all non-negative integers are eigenvalues. Hence,

$$\boxed{\lambda = 0, 1, 2, \ldots}$$

End of lecture 2

From calculations of norms above, we can choose normalised eigenstates $|n\rangle$ related by

$$a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle, \qquad a|n\rangle = \sqrt{n}\,|n-1\rangle.$$

Starting from any state $|n\rangle$ we can generate all others. In particular, starting from the ground state $|0\rangle$ characterised by

$$a|0\rangle = 0,$$

have normalised eigenstate

$$|n\rangle = \frac{1}{\sqrt{n!}}(a^\dagger)^n|0\rangle.$$

a, $a^\dagger$ also called <u>ladder operators</u>.

(b) <u>Importance of the oscillator</u>

- A smooth potential $V(x)$ with equilibrium point $x = x_0$ (so that $V'(x_0) = 0$) can be expanded

  $$V(x_0 + x) = V(x_0) + \frac{1}{2}V''(x_0)x^2 + O(x^3).$$

  So, if $x = x_0$ is <u>stable</u>, can set $V''(x_0) = m\omega^2$ & for displacements x not too large, expect system to be well-approximated by oscillator.

  e.g. diatomic molecules — quantisation of vibrational energies important in understanding internal energy & heat capacities — macroscopic consequences.

- More complicated systems can be analysed in terms of <u>normal modes</u>: coherent motion in which all degrees of freedom oscillate with same common frequency ω. General solution for <u>classical</u> oscillator is

  $$x(t) = Ae^{-i\omega t} + A^* e^{+i\omega t} \qquad A \text{ complex constant.}$$

Normal modes for a system with variables x_N are solutions of form

$$x_N(t) = Au_N e^{-i\omega t} + A^* u_N^* e^{+i\omega t}$$

with u_N complex vector specific to each normal frequency ω.

e.g. crystal with # atom $\mathcal{N} \sim 10^{23}$ — forces approximately elastic & N labels directions of motion for each atom, so $3 \times \mathcal{N}$ values.

or EM field — normal modes oscillations of electric & magnetic fields at each point in space

$$\underline{E}(\underline{x}, t) = A\underline{u}(\underline{x})e^{-i\omega t} + A^*\underline{u}(\underline{x})^* e^{+i\omega t}$$

for some complex vector field $\underline{u}(\underline{x})$. This is exact.

- QM: normal modes are independent oscillators — treat these as above. For each mode have

$$a(\omega) \quad \& \quad a(\omega)^\dagger.$$

Ground state $|0\rangle$ with

$$a(\omega)|0\rangle = 0.$$

Other states e.g. $a^\dagger(\omega)|0\rangle$ & find they have energy & momentum & behave like particles.

$$\left.\begin{matrix} a \text{ destroys} \\ a^\dagger \text{ creates} \end{matrix}\right\} \text{particles} \begin{cases} \text{phonons in crystal} \\ \text{photons in EM field} \end{cases}$$

1.3 Bases & Measurements

(a) Bases of Eigenstates

From key properties (i), (ii), (iii) in § 1.1 we know that for any observable Q there is an orthonormal basis of eigenstates $\{|n\rangle\}$ for V

$$Q|n\rangle = q_n|n\rangle, \qquad \langle m|n\rangle = \delta_{mn}.$$

For any state,

$$|\psi\rangle = \sum_n \alpha_n |n\rangle, \qquad \text{where} \quad \alpha_n = \langle n|\psi\rangle.$$

Note

$$\big\| \, |\psi\rangle \, \big\|^2 = 1 \quad \Longleftrightarrow \quad \begin{aligned} \langle\psi|\psi\rangle &= \left(\sum_m \alpha_m^* \langle m|\right)\left(\sum_n \alpha_n |n\rangle\right) \\ &= \sum_n |\alpha_n|^2 = 1. \end{aligned}$$

Note that we can write

$$1 = \sum_n |n\rangle\langle n| \qquad \text{“completeness relation”}$$

since on any state

$$\begin{aligned} |\psi\rangle &= \left(\sum_n |n\rangle\langle n|\right)|\psi\rangle \\ &= \sum_n |n\rangle\big(\langle n|\psi\rangle\big), \end{aligned}$$

so holds for any orthonormal basis.

In above there may be several independent states with same eigenvalue λ. The eigenspace for a given eval is

$$V_\lambda = \big\{|\psi\rangle : Q|\psi\rangle = \lambda|\psi\rangle\big\}.$$

& note V_λ has basis $\big\{|n\rangle : q_n = \lambda\big\}$. The degeneracy of λ is the number of states in this basis, i.e. $\dim V_\lambda$, but note that λ is also called non-degenerate if degeneracy is 1.

End of lecture 3

(b) Functions of Operators

It is straightforward to define $f(Q)$ for some operator Q if f is a polynomial or power series (assuming convergence), e.g.

$$f(Q) = Q^3 + Q \qquad \text{or} \qquad f(Q) = \exp Q = \sum_{n=0}^{\infty} \frac{1}{n!} Q^n.$$

Important examples are

$$e^{-iHt/\hbar} \qquad \text{time evolution operator } (H \text{ Hamiltonian})$$

&

$$e^{-ia\hat{p}/\hbar} \qquad \text{translation operator (see later section of Chapter 1).}$$

More generally, for Q an observable with basis of eigenstates as in (a), we can define

$$f(Q)|n\rangle = f(q_n)|n\rangle$$

& then defined on any state, provided $f(q_n)$ makes sense, e.g.

$$\frac{1}{Q}|n\rangle = \frac{1}{q_n}|n\rangle \qquad \text{defined for } q_n \neq 0.$$

In addition, note that

$$Q = \sum_n q_n |n\rangle\langle n| \qquad \& \qquad f(Q) = \sum_n f(q_n)|n\rangle\langle n|.$$

Recall, introduced eigenspaces

$$V_\lambda = \big\{|\psi\rangle : Q|\psi\rangle = \lambda|\psi\rangle\big\}.$$

A projection operator P_λ onto this subspace, $V \to V_\lambda$, can be defined by its action on basis states:

$$P_\lambda|n\rangle = \begin{cases} |n\rangle & \text{if } q_n = \lambda, \\ 0 & \text{else.} \end{cases}$$

This can be written

$$P_\lambda = \sum_{n:\, q_n = \lambda} |n\rangle\langle n|.$$

(c) Measurement Axioms

Consider a measurement of Q (observable) when the system is in state $|\psi\rangle$. Then

- The result is an eigenvalue λ of Q.
- This result is obtained with probability

$$p(\lambda) = \big\|P_\lambda|\psi\rangle\big\|^2 = \sum_{n:\, q_n = \lambda} |\alpha_n|^2.$$

- Immediately after the measurement the state is

$$cP_\lambda|\psi\rangle = c\sum_{n:\, q_n=\lambda} \alpha_n|n\rangle,$$

  with c a constant for normalisation.

Often have case with λ non-degenerate ($\dim V_\lambda = 1$), eigenstate $|n\rangle$ unique (up to a phase), & then

- $p(\lambda) = |\alpha_n|^2 = |\langle n|\psi\rangle|^2$, & $\alpha_n = \langle n|\psi\rangle$ is called the amplitude.
- State after measurement is $|n\rangle$.

In general

$$\sum_\lambda p(\lambda) = \sum_n |\alpha_n|^2 = 1.$$

Example

System with space V with basis states (orthonormal)

$$|1\rangle, \quad |2\rangle, \quad |3\rangle,$$

& these are eigenstates of Q with evals

$$q_1 = q_2 = 0, \qquad q_3 = 1.$$

State of system

$$|\psi\rangle = \frac{1}{\sqrt{6}}\big(2|1\rangle + |2\rangle + |3\rangle\big).$$

Result	Probability of measuring	State afterwards
1	$\left\|\frac{1}{\sqrt{6}}\right\|^2 = \frac{1}{6}$	$\|3\rangle$
0	$\left\|\frac{2}{\sqrt{6}}\right\|^2 + \left\|\frac{1}{\sqrt{6}}\right\|^2 = \frac{5}{6}$	$\frac{1}{\sqrt{5}}\big(2\|1\rangle + \|2\rangle\big)$

Expectation value (mean) of Q in state $|\psi\rangle$ is

$$\langle Q\rangle_\psi = \langle\psi|Q|\psi\rangle = \sum_\lambda \lambda p(\lambda) = \sum_n q_n|\alpha_n|^2,$$

& the uncertainty (spread or variance) is

$$(\Delta Q)^2_\psi = \big\langle (Q - \langle Q\rangle_\psi)^2\big\rangle_\psi = \langle Q^2\rangle_\psi - \langle Q\rangle^2_\psi.$$

In thinking about many repeated measurements: must prepare the system in the same state $|\psi\rangle$ each time.

Note: If $|\psi\rangle$ is an eigenstate of Q with eval λ, then

$$\langle Q\rangle_\psi = \lambda, \qquad \& \qquad (\Delta Q)_\psi = 0.$$

QM behaviour arises from fact that certain observables do not commute. In general, in any state $|\psi\rangle$,

$$(\Delta A)_\psi(\Delta B)_\psi \geq \frac{1}{2}\big|\langle [A,B]\rangle_\psi\big|,$$

so $[A, B] \neq 0 \Rightarrow$ cannot expect to measure A & B precisely at same time.
[Result follows from

$$\left\| (A + i\lambda B)|\psi\rangle \right\|^2 \geq 0$$

for all real λ; expand & look at discriminant of quadratic in λ.]

Prototype case: position & momentum $\hat{x}$, $\hat{p}$ obey

$$[\hat{x}, \hat{p}] = i\hbar \qquad \text{in one dimension,} \qquad \& \qquad \Delta x \, \Delta p \geq \hbar/2.$$

In three dimensions have $\hat{x}_i$ & $\hat{p}_i$ with

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij},$$

& uncertainty applies to position/momentum in a given direction.

(d) Simultaneous measurement & complete commuting sets

If observable Q does not distinguish completely, because some evals are degenerate, then we can look for an additional observable Q' such that Q, Q' are simultaneously measurable. This means we have joint eigenstates $|\lambda, \lambda'\rangle$ with

$$Q|\lambda, \lambda'\rangle = \lambda|\lambda, \lambda'\rangle,$$
$$Q'|\lambda, \lambda'\rangle = \lambda'|\lambda, \lambda'\rangle.$$

This implies

$$QQ'|\lambda, \lambda'\rangle = Q'Q|\lambda, \lambda'\rangle.$$

Hence, if we have a basis of such eigenstates, then $QQ' = Q'Q$ on any state, or $[Q, Q'] = 0$. Conversely, if $[Q, Q'] = 0$, easy to check that

$$Q' : V_\lambda \to V_\lambda \qquad (V_\lambda \text{ eigenspace for } Q),$$

& finding a basis for V_λ consisting of eigenstates of Q' then gives us a basis of joint eigenstates.

End of lecture 4

§ 1.3 (d) — continued

A set of observables $\{Q, Q', Q'', \ldots\}$ is a complete commuting set if any pair commute $\Rightarrow$ a basis of joint eigenstates, & the set of eigenvalues, $\lambda, \lambda', \lambda'', \ldots$ label states uniquely (up to normalisation).

Equivalent, if A is any observable such that

$$[A, Q] = [A, Q'] = \cdots = 0$$

then

$$A = f(Q, Q', \ldots).$$

Example

Refer to example in (c): orthonormal states $|1\rangle$, $|2\rangle$, $|3\rangle$ eigenstates of Q with $\lambda(= q_1 = q_2) = 0$, $\lambda(= q_3) = 1$.

Second observable Q' with

$$Q'|1\rangle = |2\rangle,$$
$$Q'|2\rangle = |1\rangle,$$
$$Q'|3\rangle = 0.$$

Check

$$[Q, Q'] = 0 \qquad (Q' : V_{\lambda=0} \to V_{\lambda=0}).$$

Let

$$|\pm\rangle = \frac{1}{\sqrt{2}}\big(|1\rangle \pm |2\rangle\big)$$

then

$$Q'|\pm\rangle = \pm|\pm\rangle.$$

Q, Q' complete commuting set with basis of joint eigenstates

	λ	λ'
$\|+\rangle$	0	1
$\|-\rangle$	0	-1
$\|3\rangle$	1	0

1.4 Bases, Representations & Wavefunctions

(a) Discrete & Continuous Bases

Consider orthonormal basis $\{|n\rangle\}$ with n discrete label. We can regard all states & operators as row/col vecs & matrices, respectively:

$$|\psi\rangle = \sum_n \alpha_n |n\rangle \quad \Longleftrightarrow \quad \alpha_n = \langle n|\psi\rangle,$$
$$|\phi\rangle = \sum_n \beta_n |n\rangle \quad \Longleftrightarrow \quad \beta_n = \langle n|\phi\rangle.$$

cmpts of complex vectors

Inner product

$$\langle\phi|\psi\rangle = \sum_n \beta_n^* \alpha_n.$$

For an operator A,

$$A|n\rangle = \sum_m |m\rangle A_{mn}, \qquad \text{complex matrix } A_{mn} = \langle m|A|n\rangle.$$

Simple to check that

$$|\phi\rangle = A|\psi\rangle \quad \Longleftrightarrow \quad \beta_m = \sum_n A_{mn}\alpha_n$$

& also, if B is another operator, matrix $B_{mn} = \langle m|B|n\rangle$, then

$$(AB)_{mn} = \sum_p A_{mp}B_{pn} \qquad \text{(matrix multiplication)}.$$

The choice of basis provides a representation of the quantum states & operators.

We can also allow the index n to be continuous. This requires modifications

$$\sum_n \longrightarrow \int dn$$

$$\langle n|m\rangle = \delta_{nm} \longrightarrow \delta(n-m)$$

$$|\psi\rangle = \sum_n \alpha_n|n\rangle \longrightarrow \int dn\, \alpha_n|n\rangle$$

$$\alpha_n = \langle n|\psi\rangle.$$

Completeness relation

$$1 = \sum_n |n\rangle\langle n| \longrightarrow \int dn\, |n\rangle\langle n|.$$

Note: if basis states above arise as eigenstates of observable, then

$$|\alpha_n|^2 = |\langle n|\psi\rangle|^2 \longrightarrow \text{ probability density for continuous range of } n.$$

(b) Position & Momentum Bases — Wavefunctions

Consider a particle in one dimension, position $\hat{x}$ & momentum $\hat{p}$ obeying

$$[\hat{x}, \hat{p}] = i\hbar.$$

Let $|x\rangle$ be position eigenstates

$$\hat{x}|x\rangle = x|x\rangle$$

continuous evals with

$$\langle x|x'\rangle = \delta(x - x'), \qquad \int dx\, |x\rangle\langle x| = 1.$$

Any state can be written

$$|\psi\rangle = \int dx\, \psi(x)|x\rangle$$

with

$$\psi(x) = \langle x|\psi\rangle$$

& this is the usual position space wavefunction. From remarks in (a), $|\psi(x)|^2$ is a probability density for measuring position.

Inner product

$$\begin{aligned}\langle\phi|\psi\rangle &= \left(\int dx\, \phi(x)^*\langle x|\right)\left(\int dx'\, \psi(x')|x'\rangle\right) \\ &= \int dx \int dx'\, \phi(x)^*\psi(x') \underbrace{\langle x|x'\rangle}_{\delta(x-x')} = \int dx\, \phi(x)^*\psi(x)\end{aligned}$$

&

$$\big\|\, |\psi\rangle \,\big\|^2 = \langle\psi|\psi\rangle = \int dx\, |\psi(x)|^2 = 1, \quad \text{if normalised.}$$

Similarly, define momentum eigenstates $|p\rangle$ with

$$\hat{p}|p\rangle = p|p\rangle$$

&

$$\langle p|p'\rangle = \delta(p - p'), \qquad \int dp\, |p\rangle\langle p| = 1.$$

Now can expand

$$|\psi\rangle = \int dp\, \tilde{\psi}(p)|p\rangle$$

with

$$\tilde{\psi}(p) = \langle p|\psi\rangle$$

momentum space wavefunction & $|\tilde{\psi}(p)|^2$ probability distribution for measurements of momentum.

To proceed, we need

$$\left.\begin{aligned}\langle x|p\rangle &= \frac{1}{(2\pi\hbar)^{1/2}} e^{ipx/\hbar}, \\ \langle p|x\rangle &= \frac{1}{(2\pi\hbar)^{1/2}} e^{-ipx/\hbar}.\end{aligned}\right\} \qquad (*)$$

Assume $(*)$ holds & verify it below.

Consider action of $\hat{x}$ & $\hat{p}$ on position space wavefunctions

$$\hat{x}|\psi\rangle \quad \text{wavefn}: \quad \langle x|\hat{x}|\psi\rangle = x\langle x|\psi\rangle = x\psi(x).$$

$$\begin{aligned}\hat{p}|\psi\rangle \quad \text{wavefn}: \quad & \langle x|\hat{p}|\psi\rangle \\ & = \int dp\, \langle x|\hat{p}|p\rangle\langle p|\psi\rangle \qquad \text{completeness} \\ & = \int dp\, p\langle x|p\rangle\langle p|\psi\rangle.\end{aligned}$$

Note

$$p\langle x|p\rangle = -i\hbar\frac{\partial}{\partial x}\langle x|p\rangle \qquad \text{from } (*).$$

Hence

$$\begin{aligned}\langle x|\hat{p}|\psi\rangle &= -i\hbar\frac{\partial}{\partial x}\int dp\, \langle x|p\rangle\langle p|\psi\rangle \\ &= -i\hbar\frac{\partial}{\partial x}\langle x|\psi\rangle \qquad \text{completeness again} \\ &= -i\hbar\psi'(x).\end{aligned}$$

i.e.

$$\left.\begin{aligned}\hat{x} &\to x \\ \hat{p} &\to -i\hbar\,\partial/\partial x\end{aligned}\right\} \quad \text{on position space wavefunctions.}$$

Similarly, on momentum space wavefunctions, calculate

$$\begin{matrix}\langle p|\hat{x}|\psi\rangle \\ \langle p|\hat{p}|\psi\rangle\end{matrix} \qquad \text{\& find} \qquad \begin{matrix}\hat{x} \to i\hbar\dfrac{\partial}{\partial p} \\ \hat{p} \to p.\end{matrix}$$

Relationship between wavefunctions $\psi(x)$ & $\tilde{\psi}(p)$ also follows from $(*)$

$$\begin{aligned}\tilde{\psi}(p) = \langle p|\psi\rangle &= \int dx\, \langle p|x\rangle\langle x|\psi\rangle \\ &= \frac{1}{(2\pi\hbar)^{1/2}}\int dx\, e^{-ipx/\hbar}\psi(x)\end{aligned}$$

&

$$\begin{aligned}\psi(x) = \langle x|\psi\rangle &= \int dp\, \langle x|p\rangle\langle p|\psi\rangle \\ &= \frac{1}{(2\pi\hbar)^{1/2}}\int dp\, e^{ixp/\hbar}\tilde{\psi}(p).\end{aligned}$$

Fourier transform & inverse.

End of lecture 5

Now given $H(\hat{x},\hat{p})$ we can consider

$$H|\psi\rangle = E|\psi\rangle.$$

If

$$H = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

$$\text{on } \psi(x) \qquad H \to -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x),$$
$$\text{on } \tilde{\psi}(p) \qquad H \to \frac{p^2}{2m} + V\left(i\hbar\frac{\partial}{\partial p}\right).$$

It may be easy to interpret potential term in momentum space. e.g. $V(x) = \lambda x^n$ then

$$V\left(i\hbar\frac{\partial}{\partial p}\right) = \lambda(i\hbar)^n\left(\frac{\partial}{\partial p}\right)^n$$

for n a positive integer. More generally, argue from first principles as follows.

Consider

$$\begin{aligned}
\langle p|V(\hat{x})|\psi\rangle &= \int dx\, \langle p|V(\hat{x})|x\rangle\langle x|\psi\rangle \\
&= \int dx \int dp'\, \langle p|V(\hat{x})|x\rangle\langle x|p'\rangle\langle p'|\psi\rangle \\
&= \int dp' \left(\int dx\, V(x) \underbrace{\langle p|x\rangle\langle x|p'\rangle}_{\text{use } (*)}\langle p'|\psi\rangle \right) \\
&= \int dp' \int dx\, V(x) \frac{e^{-i(p-p')x/\hbar}}{2\pi\hbar}\tilde{\psi}(p') \\
&= \frac{1}{(2\pi\hbar)^{1/2}} \int dp'\, \tilde{V}(p-p')\tilde{\psi}(p'),
\end{aligned}$$

reproducing convolution theorem.

i.e. $H|\psi\rangle = E|\psi\rangle$ becomes

$$-\frac{\hbar^2}{2m}\psi''(x) + V(x)\psi(x) = E\psi \qquad \text{in position space}$$

or

$$\frac{p^2}{2m}\tilde{\psi}(p) + \frac{1}{(2\pi\hbar)^{1/2}} \int dp'\, \tilde{V}(p-p')\tilde{\psi}(p') = E\tilde{\psi}(p)$$

in momentum space.

To generalise from one to three dimensions: position & momentum operators obey

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}.$$

One complete commuting set is

$$\underline{\hat{x}} = (\hat{x}_1, \hat{x}_2, \hat{x}_3)$$

with joint eigenstates

$$\hat{x}_i|\underline{x}\rangle = x_i|\underline{x}\rangle$$

orthonormal

$$\langle \underline{x}|\underline{x}'\rangle = \delta^{(3)}(\underline{x} - \underline{x}').$$

An alternative set

$$\underline{\hat{p}} = (\hat{p}_1, \hat{p}_2, \hat{p}_3)$$

with joint eigenstates

$$\hat{p}_i|\underline{p}\rangle = p_i|\underline{p}\rangle \qquad \text{(also orthonormal)}.$$

For any state $|\psi\rangle$ we can expand

$$|\psi\rangle = \int d^3\underline{x}\, \psi(\underline{x})|\underline{x}\rangle = \int d^3\underline{p}\, \tilde{\psi}(\underline{p})|\underline{p}\rangle$$

&

$$\psi(\underline{x}) = \langle \underline{x}|\psi\rangle, \qquad \tilde{\psi}(\underline{p}) = \langle \underline{p}|\psi\rangle.$$

$(*)$ becomes

$$\langle \underline{x}|\underline{p}\rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\underline{p}\cdot\underline{x}/\hbar}$$

& wavefunctions related by

$$\tilde{\psi}(\underline{p}) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3\underline{x}\, e^{-i\underline{p}\cdot\underline{x}/\hbar}\psi(\underline{x}) \qquad (\& \text{ inverse}).$$

(c) More on position & momentum eigenstates

Return to one dimension & seek to justify $(*)$ using only $[\hat{x},\hat{p}] = i\hbar$.

Given $|x\rangle$ with $\hat{x}|x\rangle = x|x\rangle$, consider

$$T(a) = e^{-ia\hat{p}/\hbar} = \sum_{n=0}^{\infty} \frac{1}{n!}\left(\frac{-ia}{\hbar}\right)^n \hat{p}^n \qquad a \text{ any real number}$$

& note

$$[\hat{x}, T(a)] = \left[\hat{x}, \frac{-ia\hat{p}}{\hbar}\right] T(a) = aT(a) \qquad [\text{see 3 on Example Sheet 1}].$$

Hence

$$\begin{aligned}\hat{x}\big(T(a)|x\rangle\big) &= \big([\hat{x}, T(a)] + T(a)\hat{x}\big)|x\rangle \\ &= (x+a)\big(T(a)|x\rangle\big).\end{aligned}$$

Think of $T(a)|x\rangle$ as a translated eigenstate.

Similarly, consider

$$S(b) = e^{ib\hat{x}/\hbar}$$

& note

$$[\hat{p}, S(b)] = bS(b).$$

Hence

$$\hat{p}|p\rangle = p|p\rangle \quad \Rightarrow \quad \hat{p}\big(S(b)|p\rangle\big) = (p+b)\big(S(b)|p\rangle\big).$$

$S(b)$ gives translation in momentum space.

Now take fixed values x_0 & p_0 & define all eigenstates by

$$|x_0 + a\rangle = T(a)|x_0\rangle, \qquad |p_0 + b\rangle = S(b)|p_0\rangle.$$

Now observe

$$\begin{aligned}\langle x_0 + a|p_0 + b\rangle &= \langle x_0|e^{ia\hat{p}/\hbar}|p_0 + b\rangle \\ &= \langle x_0|e^{ia(p_0+b)/\hbar}e^{ib\hat{x}/\hbar}|p_0\rangle \\ &= e^{i(ab+ap_0+bx_0)/\hbar}\underbrace{\langle x_0|p_0\rangle}_{\text{choose } = \frac{1}{\sqrt{2\pi\hbar}}}.\end{aligned}$$

for correct, consistent normalisation.

Then set $x_0 = p_0 = 0$ to get

$$\langle a|b\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{iab/\hbar} \qquad \text{which is } (*).$$

2. Evolution in Time & Pictures

2.1 The Schrödinger Equation & the Schrödinger Picture

The evolution in time of a state $|\psi(t)\rangle$ is governed (in absence of external influences or measurements) by the Schrödinger equation (SE)

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = H|\psi(t)\rangle$$

where $H = H^\dagger$ is the Hamiltonian.

Equivalently

$$-i\hbar\frac{\partial}{\partial t}\langle\psi(t)| = \langle\psi(t)|H.$$

This implies

$$\frac{\partial}{\partial t}\big(\langle\psi(t)|\psi(t)\rangle\big) = 0$$

so that normalisation of $|\psi(t)\rangle$ is preserved in time.

Solution of SE can be written

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle$$

where

$U(t) = e^{-iHt/\hbar}$ time evolution operator, assuming H independent of t.

Check immediately from

$$i\hbar\frac{\partial U(t)}{\partial t} = HU(t).$$

Recall that time-evolution problems can also be conveniently solved by working with basis of energy eigenstates $\{|n\rangle\}$ with

$$H|n\rangle = E_n|n\rangle$$

with

$e^{-iE_nt/\hbar}|n\rangle$ solution of (t-dep) SE.

End of lecture 6

§ 2.1 Sch. eqn & Sch picture

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = H|\psi\rangle$$
$$|\psi(t)\rangle = U(t)|\psi(0)\rangle$$

where

$$U(t) = e^{-iHt/\hbar} \qquad (H \text{ indep of } t)$$
$$= \sum_{n=0}^{\infty}\frac{1}{n!}\left(-\frac{it}{\hbar}H\right)^n$$

If

$$|\psi(0)\rangle = \sum_n \alpha_n|n\rangle$$

where

$$H|n\rangle = E_n|n\rangle$$

then

$$|\psi(t)\rangle = \sum_n \alpha_n e^{-iE_nt/\hbar}|n\rangle$$

Example system with two energy eigenstates

$$\begin{array}{cc} |1\rangle & |2\rangle \\ E_1 & E_2 \end{array}$$

Consider measuring Q defined by

$$Q|1\rangle = |2\rangle, \qquad Q|2\rangle = |1\rangle$$

So

$$[Q, H] \neq 0$$

Eigenstates of Q:

$$|\pm\rangle = \frac{1}{\sqrt{2}}\big(|1\rangle \pm |2\rangle\big) \qquad \text{evals } q_\pm = \pm 1$$

Take initial state $|\psi(0)\rangle = |+\rangle$

$$\Rightarrow \qquad |\psi(t)\rangle = \frac{1}{\sqrt{2}}\left(e^{-iE_1 t/\hbar}|1\rangle + e^{-iE_2 t/\hbar}|2\rangle\right)$$

Probability of measuring Q at time t & getting ± 1 is

$$\begin{aligned} &|\langle \pm|\psi(t)\rangle|^2 \\ &= \left|\frac{1}{2}(\langle 1| \pm \langle 2|)\left(e^{-iE_1 t/\hbar}|1\rangle + e^{-iE_2 t/\hbar}|2\rangle\right)\right|^2 \\ &= \left|\frac{1}{2}\left(e^{-iE_1 t/\hbar} \pm e^{-iE_2 t/\hbar}\right)\right|^2 = \begin{cases} \cos^2\big((E_1 - E_2)t/2\hbar\big) \\ \sin^2\big((E_1 - E_2)t/2\hbar\big) \end{cases} \end{aligned}$$

In basis

$$\begin{aligned} |1\rangle &\to \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ |2\rangle &\to \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{aligned} \qquad H \to \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix}$$

$$U = e^{-iHt/\hbar} \quad \to \quad \begin{pmatrix} e^{-iE_1 t/\hbar} & 0 \\ 0 & e^{-iE_2 t/\hbar} \end{pmatrix}$$

In alternative basis

$$|+\rangle \to \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|-\rangle \to \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$H|\pm\rangle = \frac{1}{\sqrt{2}}\big(E_1|1\rangle + E_2|2\rangle\big)$$

$$\left.\begin{aligned} \langle \pm|H|\pm\rangle &= \frac{1}{2}(E_1 + E_2) = \mathcal{E} \\ \langle \mp|H|\pm\rangle &= \frac{1}{2}(E_1 - E_2) = \hbar\lambda \end{aligned}\right\} \quad \text{def } \mathcal{E} \text{ \& } \lambda \text{ consts.}$$

$$H \to \begin{pmatrix} \mathcal{E} & \hbar\lambda \\ \hbar\lambda & \mathcal{E} \end{pmatrix} = \mathcal{E} I + \hbar\lambda M$$

$$I \quad 2 \times 2 \text{ identity} \qquad \& \qquad M = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Now

$$U(t) = \exp(-iHt/\hbar) \to e^{-\frac{i}{\hbar}t(\mathcal{E}I+\lambda\hbar M)}$$
$$= e^{-\frac{i}{\hbar}t\mathcal{E}}e^{-it\lambda M}$$

Note $M^2 = I$ & hence

$$\exp(-it\lambda M) = \sum_{r=0}^{\infty}\frac{1}{(2r)!}(-i\lambda t)^{2r}I + \sum_{r=0}^{\infty}\frac{1}{(2r+1)!}(-i\lambda t)^{2r+1}M$$
$$= (\cos\lambda t)I - i(\sin\lambda t)M.$$

$$|\psi(0)\rangle = |+\rangle \to \begin{pmatrix}1\\0\end{pmatrix}$$

$$|\psi(t)\rangle = e^{-iHt/\hbar}|+\rangle \to \left(e^{-it\mathcal{E}/\hbar}\right)\begin{pmatrix}\cos\lambda t\\-i\sin\lambda t\end{pmatrix}$$

$$e^{-it\mathcal{E}/\hbar}\big(\cos\lambda t\,|+\rangle - i\sin\lambda t\,|-\rangle\big) \qquad \longleftarrow$$

Now compare with previous results for prob. of measuring Q & finding ± 1

2.2 Heisenberg Picture & Heisenberg Equations

(a) Unitary operators

Physical predictions in QM are given by probability amplitudes, eigenvalues, expectation values, etc., hence by expressions like

$$\langle\phi|\psi\rangle, \quad \langle\phi|A|\psi\rangle, \quad \text{etc.}$$

An operator U is called unitary if

$$UU^\dagger = U^\dagger U = 1 \quad \text{or} \quad U^\dagger = U^{-1}.$$

Given any such U, consider a map on states & operators

$$|\psi\rangle \longrightarrow |\psi'\rangle = U|\psi\rangle$$
$$\langle\psi| \longrightarrow \langle\psi'| = \langle\psi|U^\dagger$$
$$A \longrightarrow A' = UAU^\dagger$$

Under this, all physical predictions remain unchanged

$$\text{e.g.} \qquad \langle\phi|\psi\rangle \to \langle\phi'|\psi'\rangle = \langle\phi|U^\dagger U|\psi\rangle = \langle\phi|\psi\rangle$$
$$\langle\phi|A|\psi\rangle \to \langle\phi'|A'|\psi'\rangle = \langle\phi|U^\dagger UAU^\dagger U|\psi\rangle$$
$$= \langle\phi|A|\psi\rangle.$$

Furthermore

$$C = AB \longrightarrow C' = A'B'$$
$$C = [A,B] \longrightarrow C' = [A',B']$$
$$Q^\dagger = Q \quad \Rightarrow \quad Q' = UQU^\dagger \quad \text{also satisfies } (Q')^\dagger = Q'$$
$$Q|\psi\rangle = \lambda|\psi\rangle \quad \Rightarrow \quad Q'|\psi'\rangle = \lambda|\psi'\rangle$$
$$\text{same eigenvalue.}$$

(b) Changing Pictures

In the Schrödinger picture

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = H|\psi(t)\rangle \qquad \begin{pmatrix}\text{all for } H \text{ indep of } t \\ \& \ H = H^\dagger\end{pmatrix}$$

$$\Rightarrow \qquad |\psi(t)\rangle = U(t)|\psi(0)\rangle$$

where $U(t) = \exp(-iHt/\hbar)$ is unitary. In fact

$$U(t)^\dagger = U(t)^{-1} = U(-t)$$

$$\& \qquad U(t_1)U(t_2) = U(t_1 + t_2)$$

The states depend on t but observables do not, in S-picture.

To pass to Heisenberg picture we shift t-dependence from states to operators as follows

	Schrödinger	Heisenberg
states	$\|\psi(t)\rangle_S$	$\|\psi\rangle_H = U^\dagger\|\psi(t)\rangle_S = e^{itH/\hbar}\|\psi(t)\rangle_S = \|\psi(0)\rangle_S$
ops	A_S	$A_H(t) = U^\dagger A_S U = e^{itH/\hbar}A_S e^{-itH/\hbar}$

Because the transformation $U(t)$ is unitary

$${}_S\langle\phi(t)|A_S|\psi(t)\rangle_S = {}_H\langle\phi|A_H(t)|\psi\rangle_H$$

Note

$$H_H = H_S = H$$

To specify dynamics in H-picture, consider

$$\begin{aligned}\frac{d}{dt}A_H(t) &= \frac{d}{dt}\left(e^{iHt/\hbar}A_S e^{-iHt/\hbar}\right) \\ &= \frac{iH}{\hbar}e^{iHt/\hbar}A_S e^{-iHt/\hbar} - e^{iHt/\hbar}A_S e^{-iHt/\hbar}\frac{iH}{\hbar} \\ &= \frac{i}{\hbar}[H, A_H(t)]\end{aligned}$$

or

$$i\hbar\frac{d}{dt}A_H(t) = [A_H(t), H] \qquad \text{Heisenberg equations of motion}$$

e.g. particle in one dimension $\hat{x}(t)$, $\hat{p}(t)$ in H-picture

consider

$$H = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

$$\frac{d}{dt}\hat{x}(t) = \frac{1}{i\hbar}[\hat{x}(t), H] = \frac{1}{m}\hat{p}(t)$$

$$\frac{d}{dt}\hat{p}(t) = \frac{1}{i\hbar}[\hat{p}(t), H] = -V'(\hat{x}(t))$$

using

$$[\hat{x}(t), \hat{p}(t)] = i\hbar$$

End of lecture 7

Taking expectation values in any state $|\psi\rangle$ (independent of t in the Heisenberg picture) gives

$$\frac{d}{dt}\langle\hat{x}\rangle = \frac{1}{m}\langle\hat{p}\rangle, \qquad \frac{d}{dt}\langle\hat{p}\rangle = -\langle V'(\hat{x})\rangle.$$

These are Ehrenfest's theorem, true in all pictures.

For some potentials we can solve the Heisenberg equations explicitly.

Free particle: $V = 0$

$$\frac{d}{dt}\hat{p}(t) = 0 \quad \Rightarrow \quad \hat{p}(t) = \hat{p}(0) \qquad \text{(constant operator)}.$$

Then

$$\frac{d}{dt}\hat{x}(t) = \frac{1}{m}\hat{p}(0) \quad \Rightarrow \quad \hat{x}(t) = \hat{x}(0) + \frac{\hat{p}(0)}{m}t,$$

just as in classical dynamics.

Harmonic oscillator: $V = \frac{1}{2}m\omega^2 x^2$

$$\frac{d}{dt}\hat{x}(t) = \frac{1}{m}\hat{p}(t), \qquad \frac{d}{dt}\hat{p}(t) = -m\omega^2\hat{x}(t).$$

Hence

$$\frac{d^2}{dt^2}\hat{x}(t) + \omega^2\hat{x}(t) = 0, \qquad \frac{d^2}{dt^2}\hat{p}(t) + \omega^2\hat{p}(t) = 0.$$

The solutions are

$$\hat{x}(t) = \hat{x}(0)\cos\omega t + \frac{\hat{p}(0)}{m\omega}\sin\omega t,$$
$$\hat{p}(t) = \hat{p}(0)\cos\omega t - m\omega\hat{x}(0)\sin\omega t.$$

Or equivalently,

$$\hat{x}(t) = \left(\frac{\hbar}{2m\omega}\right)^{1/2}\left(ae^{-i\omega t} + a^\dagger e^{i\omega t}\right),$$
$$\hat{p}(t) = \left(\frac{\hbar m\omega}{2}\right)^{1/2}\frac{1}{i}\left(ae^{-i\omega t} - a^\dagger e^{i\omega t}\right),$$

where a, $a^\dagger$ are defined from the Schrödinger-picture operators $\hat{x}(0)$, $\hat{p}(0)$.

Why do the ladder operators acquire these phases?

For the harmonic oscillator,

$$H = \hbar\omega\left(a^\dagger a + \frac{1}{2}\right), \qquad [a, a^\dagger] = 1.$$

Using $[AB, C] = A[B, C] + [A, C]B$, we find

$$[a^\dagger a, a] = a^\dagger[a, a] + [a^\dagger, a]a = -a,$$
$$[a^\dagger a, a^\dagger] = a^\dagger[a, a^\dagger] + [a^\dagger, a^\dagger]a = a^\dagger.$$

The constant term in H commutes with every operator, so

$$[H, a] = -\hbar\omega a, \qquad [H, a^\dagger] = \hbar\omega a^\dagger.$$

Define

$$a_H(t) = U^\dagger(t)aU(t), \qquad a_H^\dagger(t) = U^\dagger(t)a^\dagger U(t).$$

Since H commutes with $U(t)$, the same commutator relations hold for the Heisenberg-picture operators. Their equations of motion are therefore

$$\frac{d}{dt}a_H(t) = \frac{i}{\hbar}[H, a_H(t)] = -i\omega a_H(t),$$
$$\frac{d}{dt}a_H^\dagger(t) = \frac{i}{\hbar}[H, a_H^\dagger(t)] = i\omega a_H^\dagger(t).$$

With $a_H(0) = a$ and $a_H^\dagger(0) = a^\dagger$, this gives

$$\boxed{a_H(t) = ae^{-i\omega t}, \qquad a_H^\dagger(t) = a^\dagger e^{i\omega t}.}$$

The two phase factors have opposite signs because the commutators of H with the lowering and raising operators have opposite signs. The second result also follows by taking the adjoint of the first. Substituting these results into

$$\hat{x}(t) = \sqrt{\frac{\hbar}{2m\omega}}\big(a_H(t) + a_H^\dagger(t)\big), \qquad \hat{p}(t) = \frac{1}{i}\sqrt{\frac{\hbar m\omega}{2}}\big(a_H(t) - a_H^\dagger(t)\big)$$

recovers the expressions above. Here a and $a^\dagger$ without a time argument denote the fixed operators at $t = 0$.

2.3 Canonical Quantisation

$*$ Non-examinable material

This is the final step in Dirac's approach to constructing QM: how does one pass from a general classical system to its quantum version?

A classical system can be described by a set of generalised positions $x_i(t)$ and momenta $p_i(t)$, $1 \le i \le N$ (which may include angles and angular momenta). These are coordinates on phase space. The dynamics are governed by a Hamiltonian $H(x_i, p_i)$.

The Poisson bracket of two functions $f(x_i, p_i)$, $g(x_i, p_i)$ is defined by

$$\{f, g\} = \sum_{i=1}^{N}\left(\frac{\partial f}{\partial x_i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial x_i}\right),$$

which is another function on phase space. In particular,

$$\{x_i, p_j\} = \delta_{ij}.$$

Properties of Poisson brackets closely resemble those of commutators (linearity, the Jacobi identity, etc.). Dynamics in this formulation are given by Hamilton's equation

$$\frac{df}{dt} = \{f, H\} \qquad \text{for } f \text{ with no explicit time dependence.}$$

In canonical quantisation, classical functions are promoted to quantum operators,

$$f, g \quad \longrightarrow \quad \hat{f}, \hat{g},$$

and Poisson brackets determine commutators through Dirac's formal rule

$$[\hat{f}, \hat{g}] = i\hbar\widehat{\{f, g\}}.$$

In particular, this recovers

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij},$$

and the correspondence with Hamilton's equation gives the Heisenberg equation

$$i\hbar\frac{d\hat{f}}{dt} = [\hat{f}, H].$$

The rule is a quantisation prescription; for general functions, operator ordering must be specified and the rule cannot hold exactly for every pair of functions simultaneously.

$*$ End of non-examinable material

Tensor Products

Given state spaces V_1 and V_2, their tensor product $V = V_1 \otimes V_2$ is spanned by tensor product states

$$|\psi\rangle \otimes |\phi\rangle, \qquad |\psi\rangle \in V_1, \quad |\phi\rangle \in V_2,$$

with duals $\langle\psi| \otimes \langle\phi|$, subject to

$$\begin{aligned}(|\psi\rangle + |\psi'\rangle) \otimes |\phi\rangle &= |\psi\rangle \otimes |\phi\rangle + |\psi'\rangle \otimes |\phi\rangle,\\ |\psi\rangle \otimes (|\phi\rangle + |\phi'\rangle) &= |\psi\rangle \otimes |\phi\rangle + |\psi\rangle \otimes |\phi'\rangle,\\ (\alpha|\psi\rangle) \otimes |\phi\rangle &= |\psi\rangle \otimes (\alpha|\phi\rangle) = \alpha\big(|\psi\rangle \otimes |\phi\rangle\big),\end{aligned}$$

and similarly for duals.

The inner product on V satisfies

$$\big(\langle\psi'| \otimes \langle\phi'|\big)\big(|\psi\rangle \otimes |\phi\rangle\big) = \langle\psi'|\psi\rangle\langle\phi'|\phi\rangle,$$

extended sesquilinearly to arbitrary states (conjugate linear in the first state and linear in the second).

Given operators A on V_1 and B on V_2, define an operator $A \otimes B$ on V by

$$(A \otimes B)\big(|\psi\rangle \otimes |\phi\rangle\big) = (A|\psi\rangle) \otimes (B|\phi\rangle),$$

and extend by linearity. In particular,

$$A \longleftrightarrow A \otimes 1 \quad \text{acting only on } V_1, \qquad B \longleftrightarrow 1 \otimes B \quad \text{acting only on } V_2.$$

Operators acting on different factors commute for any A, B:

$$[A \otimes 1, 1 \otimes B] = 0.$$

Example: a particle in two dimensions

Consider the position operators $\hat{x}_1$, $\hat{x}_2$. A basis of joint eigenstates is constructed as

$$\underbrace{|x_1, x_2\rangle}_{\in V} = \underbrace{|x_1\rangle}_{\in V_1} \otimes \underbrace{|x_2\rangle}_{\in V_2},$$

where V_1, V_2 are the state spaces for a particle in one dimension. The position operators are identified with

$$\hat{x}_1 \longleftrightarrow \hat{x}_1 \otimes 1, \qquad \hat{x}_2 \longleftrightarrow 1 \otimes \hat{x}_2.$$

The wavefunction for a product state $|\psi\rangle \otimes |\phi\rangle$ is given by

$$\begin{aligned}\langle x_1, x_2|\big(|\psi\rangle \otimes |\phi\rangle\big) &= (\langle x_1| \otimes \langle x_2|)(|\psi\rangle \otimes |\phi\rangle)\\ &= \langle x_1|\psi\rangle\langle x_2|\phi\rangle\\ &= \psi(x_1)\phi(x_2).\end{aligned}$$

Thus

tensor product states $\longleftrightarrow$ separable wavefunctions,

tensor product space $\longleftrightarrow$ all linear combinations of such wavefunctions.

Note. If $\{|n\rangle\}$ is a basis for V_1 and $\{|m\rangle\}$ is a basis for V_2, then

$$\{|n\rangle \otimes |m\rangle\}$$

is a basis for V. If both bases are finite,

$$\dim V = (\dim V_1)(\dim V_2).$$

End of lecture 8

Example: the two-dimensional isotropic oscillator

The Hamiltonian is

$$H = \frac{1}{2m}\left(\hat{p}_1^2 + \hat{p}_2^2\right) + \frac{1}{2}m\omega^2\left(\hat{x}_1^2 + \hat{x}_2^2\right) = H_1 + H_2,$$

where

$$H_i = \frac{\hat{p}_i^2}{2m} + \frac{1}{2}m\omega^2\hat{x}_i^2 = \hbar\omega\left(N_i + \frac{1}{2}\right), \qquad i = 1, 2.$$

Here $N_i = a_i^\dagger a_i$, with

$$[a_i, a_j^\dagger] = \delta_{ij}.$$

Since $[N_1, N_2] = 0$, we construct simultaneous eigenstates

$$|n_1, n_2\rangle = |n_1\rangle \otimes |n_2\rangle, \qquad n_1, n_2 = 0, 1, 2, \ldots.$$

Then

$$\begin{aligned} H|n_1, n_2\rangle &= (H_1|n_1\rangle) \otimes |n_2\rangle + |n_1\rangle \otimes (H_2|n_2\rangle) \\ &= E_{n_1 n_2}|n_1, n_2\rangle, \end{aligned}$$

where

$$E_{n_1 n_2} = \hbar\omega(n_1 + n_2 + 1).$$

The corresponding wavefunctions factorise:

$$\begin{aligned} \psi_{n_1 n_2}(x_1, x_2) &= \langle x_1, x_2|n_1, n_2\rangle \\ &= \psi_{n_1}(x_1)\psi_{n_2}(x_2), \end{aligned}$$

where $\psi_n(x) = \langle x|n\rangle$ is the wavefunction of the one-dimensional oscillator.
Notation. It is common to suppress the tensor product symbol and write $|n_1\rangle|n_2\rangle$ for $|n_1\rangle \otimes |n_2\rangle$.

3.2 Spin

Experiment shows that particles generally carry an internal degree of freedom called spin, or intrinsic angular momentum. Even if a particle appears to be elementary or point-like, its state space has the form

$$V = V_{\text{space}} \otimes V_{\text{spin}},$$

with basis states

$$|\mathbf{x}, r\rangle = |\mathbf{x}\rangle \otimes |r\rangle,$$

where r takes a finite set of values.

The position operators alone do not, in general, form a complete commuting set: there are other observables Q acting only on V_{spin}, with

$$[\hat{x}_i, Q] = 0.$$

We return to these operators in Chapter 5; for now we focus on states.

Each kind of particle has a definite total spin s, a non-negative integer or half-integer,

$$s = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots.$$

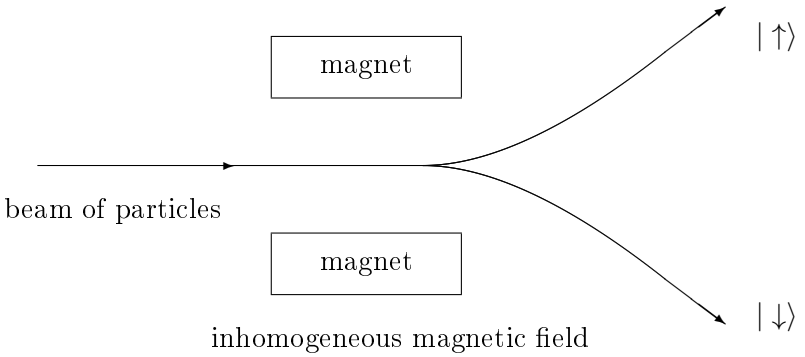
This is part of its nature, like mass or charge. For a massive particle of spin s, the spin space has $2s+1$ basis states, labelled by

$$r = s, s-1, \ldots, -s.$$

For example,

s	basis states in V_{spin}
0	$\lvert 0\rangle$
$\frac{1}{2}$	$\lvert\frac{1}{2}\rangle,\ \lvert-\frac{1}{2}\rangle$ also written $\lvert\uparrow\rangle,\ \lvert\downarrow\rangle$ (up, down)
1	$\lvert 1\rangle,\ \lvert 0\rangle,\ \lvert -1\rangle$

The existence of spin states is revealed, for example, by the Stern–Gerlach experiment. A magnet produces an inhomogeneous magnetic field, which exerts different forces on different spin states. For $s = \frac{1}{2}$, the beam splits into two:



A general state $|\psi\rangle \in V$ has a $(2s+1)$-component wavefunction

$$\psi(\mathbf{x}, r) = \langle \mathbf{x}, r|\psi\rangle.$$

Here $|\psi(\mathbf{x}, r)|^2$ is the probability density for the particle's position with spin state r. A mixed or hybrid notation is also common:

$$\sum_r \psi(\mathbf{x}, r)|r\rangle,$$

which is a spin-space-valued function of position.

3.3 Multiparticle: bosons & fermions

Consider particles labelled by $a = 1, 2, \ldots, N$. Let V_a be the state space for each particle, with basis $\{|\mathbf{x}_a, r_a\rangle\}$. General multiparticle states belong to

$$V_1 \otimes V_2 \otimes \cdots \otimes V_N,$$

with basis states

$$\begin{aligned}&|\mathbf{x}_1, r_1, \mathbf{x}_2, r_2, \ldots, \mathbf{x}_N, r_N\rangle \\ &\quad = |\mathbf{x}_1, r_1\rangle \otimes |\mathbf{x}_2, r_2\rangle \otimes \cdots \otimes |\mathbf{x}_N, r_N\rangle.\end{aligned}$$

(a) Identical particles: Bose–Einstein & Fermi–Dirac statistics

If the particles are identical, then $V_a \cong V$ for each a. Consider first $N = 2$. Define an exchange operator U by its action on basis states:

$$U|\mathbf{x}_1, r_1, \mathbf{x}_2, r_2\rangle = |\mathbf{x}_2, r_2, \mathbf{x}_1, r_1\rangle.$$

For a physical two-particle state,

$$U|\Psi\rangle = \eta|\Psi\rangle,$$

because $|\Psi\rangle$ and $U|\Psi\rangle$ are physically equivalent states. But $U^2 = 1$, as follows from its action on basis states, so

$$\eta^2 = 1, \qquad \eta = \pm 1.$$

Thus physical states lie in one of the two subspaces of $V \otimes V$:

$$\begin{aligned} (V \otimes V)_{\mathrm{S}} &= \mathrm{span}\big\{|\psi\rangle \otimes |\phi\rangle + |\phi\rangle \otimes |\psi\rangle\big\}, && \eta = +1, \\ (V \otimes V)_{\mathrm{A}} &= \mathrm{span}\big\{|\psi\rangle \otimes |\phi\rangle - |\phi\rangle \otimes |\psi\rangle\big\}, && \eta = -1, \end{aligned}$$

where $|\psi\rangle, |\phi\rangle$ range over V. These are the symmetric and antisymmetric subspaces, respectively; the displayed spanning states need not be normalised.

Similarly, for $N \geq 2$, define $U_{(a,b)}$ to exchange the labels $(\mathbf{x}_a, r_a)$ and $(\mathbf{x}_b, r_b)$ in a basis state. For a physical N-particle state,

$$U_{(a,b)}|\Psi\rangle = \eta_{(a,b)}|\Psi\rangle, \qquad \eta_{(a,b)} = \pm 1,$$

since $U_{(a,b)}^2 = 1$.

For any permutation π of $\{1, 2, \ldots, N\}$, define

$$\begin{aligned} &U_\pi|\mathbf{x}_1, r_1, \mathbf{x}_2, r_2, \ldots, \mathbf{x}_N, r_N\rangle \\ &\qquad = |\mathbf{x}_{\pi(1)}, r_{\pi(1)}, \mathbf{x}_{\pi(2)}, r_{\pi(2)}, \ldots, \mathbf{x}_{\pi(N)}, r_{\pi(N)}\rangle. \end{aligned}$$

On a physical state,

$$U_\pi|\Psi\rangle = \eta_\pi|\Psi\rangle.$$

The algebra of swaps (transpositions) requires $\eta_{(a,b)}$ to have the same value for every pair (a, b). Every permutation can be expressed as a sequence of swaps, so there are two possibilities:

$$\eta_\pi = \begin{cases} 1 & \text{for all } \pi \quad \text{(bosons)}, \\ \mathrm{sgn}(\pi) = (-1)^{\text{number of swaps in } \pi} & \text{for all } \pi \quad \text{(fermions)}. \end{cases}$$

The same alternative must be chosen for all permutations.

End of lecture 9

Bosons and fermions

- Bosons obey Bose–Einstein statistics: exchanging two identical bosons leaves the state unchanged, $\eta = +1$. Their multiparticle states belong to

$$(V \otimes V \otimes \cdots \otimes V)_{\mathrm{S}}.$$

- Fermions obey Fermi–Dirac statistics: exchanging two identical fermions changes the sign of the state, $\eta = -1$. Their multiparticle states belong to

$$(V \otimes V \otimes \cdots \otimes V)_{\mathrm{A}}.$$

These exchange conditions apply to identical particles.

(b) Spin and statistics

The spin–statistics principle states that

$$\begin{aligned} \text{integer spin } s = 0, 1, 2, \ldots \quad &\Longrightarrow \quad \text{bosons}, \\ \text{half-integer spin } s = \tfrac{1}{2}, \tfrac{3}{2}, \ldots \quad &\Longrightarrow \quad \text{fermions}. \end{aligned}$$

This is confirmed experimentally. In relativistic quantum field theory it can be derived as the spin–statistics theorem.

- Electrons, neutrinos, quarks, muons and tau leptons are spin-$\frac{1}{2}$ elementary fermions. Protons and neutrons also have spin $\frac{1}{2}$ and are fermions, although they are composite.
- The gauge bosons associated with forces have spin 1: photons (electromagnetism), $W^{\pm}$ and Z bosons (the weak interaction), and gluons (the strong interaction).
- The Higgs boson has spin 0.
- The hypothetical graviton would have spin 2.

The spin–statistics principle also applies to composite particles.

(c) Two-particle examples

For two identical particles, physical states belong to $(V \otimes V)_{\mathrm{S}}$ for bosons or $(V \otimes V)_{\mathrm{A}}$ for fermions, where

$$V = V_{\text{space}} \otimes V_{\text{spin}}.$$

It is convenient first to construct spatial and spin states with definite exchange symmetry separately, and then combine them to obtain the required symmetry of the *total* state. The exchange signs of the two factors multiply.

Consider a spin-independent one-particle Hamiltonian $H(\hat{\mathbf{x}}, \hat{\mathbf{p}})$ with non-degenerate spatial energy levels

$$E_0 < E_1 < E_2 < \cdots, \qquad \text{wavefunctions } \psi_0(\mathbf{x}), \psi_1(\mathbf{x}), \ldots.$$

For two non-interacting particles, the Hamiltonian is

$$H_{\text{tot}} = H(\hat{\mathbf{x}}_1, \hat{\mathbf{p}}_1) + H(\hat{\mathbf{x}}_2, \hat{\mathbf{p}}_2).$$

A basis of spatial energy eigenfunctions is

$$\psi_i(\mathbf{x}_1)\psi_j(\mathbf{x}_2), \qquad E = E_i + E_j.$$

The lowest spatial level is symmetric:

$$\Psi_0^{\mathrm{S}}(\mathbf{x}_1, \mathbf{x}_2) = \psi_0(\mathbf{x}_1)\psi_0(\mathbf{x}_2), \qquad E = 2E_0.$$

At the next level, the products $\psi_0(\mathbf{x}_1)\psi_1(\mathbf{x}_2)$ and $\psi_1(\mathbf{x}_1)\psi_0(\mathbf{x}_2)$ do not individually have definite exchange symmetry. Their normalised combinations do:

$$\Psi_1^{\mathrm{S}}(\mathbf{x}_1, \mathbf{x}_2) = \frac{1}{\sqrt{2}}\big(\psi_0(\mathbf{x}_1)\psi_1(\mathbf{x}_2) + \psi_1(\mathbf{x}_1)\psi_0(\mathbf{x}_2)\big),$$
$$\Psi_1^{\mathrm{A}}(\mathbf{x}_1, \mathbf{x}_2) = \frac{1}{\sqrt{2}}\big(\psi_0(\mathbf{x}_1)\psi_1(\mathbf{x}_2) - \psi_1(\mathbf{x}_1)\psi_0(\mathbf{x}_2)\big),$$

both with energy $E_0 + E_1$.

Spin 0

The spin space has only one state, so its factor can be suppressed. Spin–statistics requires bosons, hence symmetric spatial states in $(V_{\text{space}} \otimes V_{\text{space}})_{\mathrm{S}}$. The lowest two levels are therefore

spatial state	energy	degeneracy
Ψ_0^{S}	$2E_0$	1
Ψ_1^{S}	$E_0 + E_1$	1

Spin $\frac{1}{2}$

For each particle, V_{spin} has basis $\{|\uparrow\rangle, |\downarrow\rangle\}$. Thus the two-particle spin space has product basis

$$\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\},$$

where, for example, $|\uparrow\downarrow\rangle = |\uparrow\rangle \otimes |\downarrow\rangle$. A normalised basis with definite exchange symmetry consists of the three symmetric spin states

$$\chi_+ = |\uparrow\uparrow\rangle, \qquad \chi_0 = \frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}, \qquad \chi_- = |\downarrow\downarrow\rangle,$$

and the single antisymmetric spin state

$$\chi_{\mathrm{A}} = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}.$$

These are the spin triplet and singlet, respectively.

Spin–statistics requires fermions and an antisymmetric total state. After regrouping the spatial and spin tensor factors,

$$\begin{aligned}(V \otimes V)_{\mathrm{A}} \cong &\big[(V_{\mathrm{space}} \otimes V_{\mathrm{space}})_{\mathrm{A}} \otimes (V_{\mathrm{spin}} \otimes V_{\mathrm{spin}})_{\mathrm{S}}\big] \\ &\oplus \big[(V_{\mathrm{space}} \otimes V_{\mathrm{space}})_{\mathrm{S}} \otimes (V_{\mathrm{spin}} \otimes V_{\mathrm{spin}})_{\mathrm{A}}\big].\end{aligned}$$

Thus a symmetric spatial state must be paired with the spin singlet, whereas an antisymmetric spatial state must be paired with a spin triplet state. The lowest two energy levels are

total states	energy	degeneracy
$\Psi_0^{\mathrm{S}}\chi_{\mathrm{A}}$	$2E_0$	1
$\Psi_1^{\mathrm{S}}\chi_{\mathrm{A}}, \quad \Psi_1^{\mathrm{A}}\chi_m \ (m = +, 0, -)$	$E_0 + E_1$	4

3.4 Pauli Exclusion Principle & Atomic Structure

To construct a state of N identical fermions, take N one-particle states in V and antisymmetrise their tensor product. Such antisymmetrised products span $(V \otimes \cdots \otimes V)_{\mathrm{A}}$; a general state may be a linear combination of them. If two of the one-particle states are the same, the antisymmetrised product vanishes. This gives the Pauli exclusion principle: no two identical fermions can occupy the same complete one-particle state.

Why two electrons cannot occupy the same state

For two electrons, antisymmetrisation gives a state proportional to

$$|\alpha\rangle \otimes |\beta\rangle - |\beta\rangle \otimes |\alpha\rangle.$$

If $|\beta\rangle = |\alpha\rangle$, then

$$|\alpha\rangle \otimes |\alpha\rangle - |\alpha\rangle \otimes |\alpha\rangle = 0.$$

The zero vector cannot be normalised and does not represent a physical state. For N fermions, the terms in the antisymmetrisation similarly cancel in pairs whenever two occupied one-particle states coincide.

Here "the same state" means the same *complete* state in $V_{\mathrm{space}} \otimes V_{\mathrm{spin}}$, including spin. Two electrons can occupy the same spatial orbital with opposite spins: their complete one-particle states are then different, and the total state is the symmetric spatial product multiplied by χ_{A}.

Application to atomic structure

Consider N electrons bound to a nucleus containing Z protons. Ignoring electron–electron interactions and treating the nucleus as fixed, the Hamiltonian is

$$H_{\mathrm{tot}} = \sum_{a=1}^{N} H(\hat{\mathbf{x}}_a, \hat{\mathbf{p}}_a),$$

where the one-particle Hamiltonian is

$$H(\hat{\mathbf{x}}, \hat{\mathbf{p}}) = \frac{\hat{\mathbf{p}}^{\,2}}{2m} - \frac{Ze^2}{4\pi\varepsilon_0|\hat{\mathbf{x}}|}.$$

End of lecture 10

Single-electron states and shell filling

The single-electron spatial states $|n\ell m\rangle \in V_{\text{space}}$ are joint eigenstates of H, L^2 and L_3:

$$\begin{aligned} H|n\ell m\rangle &= E_n|n\ell m\rangle, \\ L^2|n\ell m\rangle &= \hbar^2\ell(\ell+1)|n\ell m\rangle, \\ L_3|n\ell m\rangle &= \hbar m|n\ell m\rangle, \end{aligned} \qquad E_n = -\frac{m}{2}\left(\frac{Ze^2}{4\pi\varepsilon_0\hbar}\right)^2\frac{1}{n^2}.$$

The allowed quantum numbers are

$$n = 1, 2, \ldots, \qquad \ell = 0, 1, \ldots, n-1, \qquad m = -\ell, -\ell+1, \ldots, \ell.$$

The integer n is the principal quantum number. For fixed ℓ there are $2\ell+1$ states, so the spatial degeneracy of E_n is

$$\sum_{\ell=0}^{n-1}(2\ell+1) = 1 + 3 + \cdots + (2n-1) = n^2.$$

Including V_{spin}, with basis $\{|\uparrow\rangle, |\downarrow\rangle\}$, gives $2n^2$ one-electron states at energy E_n.

For an electrically neutral atom, $N = Z$. Fill the one-electron levels starting with the lowest energy, subject to the Pauli exclusion principle. This gives a rough picture of the periodic table and some qualitative insight into chemical properties. In particular, completely filled shells suggest relatively unreactive elements:

filled levels	Z	element
E_1	2	He
$E_1,\ E_2$	$2 + 8 = 10$	Ne

Electron–electron interactions modify this simple model in real atoms.

4. Perturbation Theory

Few quantum-mechanical systems can be solved exactly. In perturbation theory, start from a known, solvable system

$$H|n\rangle = E_n|n\rangle,$$

where $\{|n\rangle\}$ is an orthonormal basis of energy eigenstates, and calculate the energies and eigenstates of a perturbed Hamiltonian:

$$(H + \lambda V)|\psi\rangle = E|\psi\rangle. \tag{1}$$

Here V is a Hermitian operator and λ is a parameter. Assume that E and $|\psi\rangle$ depend smoothly on λ and seek them order by order in λ. Although the discussion concerns Hamiltonians and energies, the technique applies to any observable.

4.1 Non-degenerate Case

Suppose that, as $\lambda \to 0$,

$$|\psi\rangle \to |r\rangle, \qquad E \to E_r,$$

where $|r\rangle$ is the unique eigenstate with energy E_r: the unperturbed level is non-degenerate. Using the unperturbed basis also for $\lambda \neq 0$, write

$$|\psi\rangle = \alpha|r\rangle + \sum_{j\neq r}\beta_j|j\rangle = \alpha\left(|r\rangle + \sum_{j\neq r}\gamma_j|j\rangle\right), \qquad \gamma_j = \frac{\beta_j}{\alpha}.$$

The coefficients are power series in λ, with $\alpha \to 1$ and $\beta_j, \gamma_j \to 0$ as $\lambda \to 0$. We aim to determine the coefficients in

$$\begin{aligned} E &= E_r + \lambda E_r' + \lambda^2 E_r'' + \cdots , \\ \alpha &= 1 + a\lambda + \cdots , \\ \gamma_j &= c_j \lambda + \cdots . \end{aligned}$$

Here E_r' and E_r'' denote the coefficients of λ and λ^2, respectively; E_r'' is not the second derivative.

Substitute the state expansion into (1) and cancel α:

$$(E_r + \lambda V)|r\rangle + \sum_{j\neq r} \gamma_j (E_j + \lambda V)|j\rangle = E\left(|r\rangle + \sum_{j\neq r} \gamma_j |j\rangle\right).$$

Rearranging gives the exact equation

$$(E - E_r)|r\rangle + \sum_{j\neq r} \gamma_j (E - E_j)|j\rangle = \lambda V|r\rangle + \lambda \sum_{j\neq r} \gamma_j V|j\rangle. \tag{2}$$

First-order energy correction

Multiplying (2) by $\langle r|$ gives

$$E - E_r = \lambda \langle r|V|r\rangle + \lambda \sum_{j\neq r} \gamma_j \langle r|V|j\rangle.$$

This is exact. Substituting the power series,

$$\lambda E_r' + \lambda^2 E_r'' + \cdots = \lambda \langle r|V|r\rangle + \lambda^2 \sum_{j\neq r} c_j \langle r|V|j\rangle + \cdots .$$

Equating the coefficients of λ yields

$$\boxed{E_r' = \langle r|V|r\rangle.}$$

State mixing and the second-order energy correction

To obtain the second-order correction, we need c_j. Multiplying (2) by $\langle i|$, where $i \neq r$, gives the exact relation

$$\gamma_i (E - E_i) = \lambda \langle i|V|r\rangle + \lambda \sum_{j\neq r} \gamma_j \langle i|V|j\rangle.$$

To leading order in λ,

$$\lambda c_i (E_r - E_i) = \lambda \langle i|V|r\rangle, \qquad \Longrightarrow \qquad c_i = \frac{\langle i|V|r\rangle}{E_r - E_i}.$$

The denominator is nonzero because the level E_r is non-degenerate. Thus

$$|\psi\rangle = \alpha \left[|r\rangle + \lambda \sum_{j\neq r} \frac{\langle j|V|r\rangle}{E_r - E_j} |j\rangle + O(\lambda^2) \right],$$

where α fixes the normalisation and phase. Returning to the coefficient of λ^2 in the energy equation, and using Hermiticity of V,

$$E_r'' = \sum_{j\neq r} c_j \langle r|V|j\rangle = \sum_{j\neq r} \frac{|\langle j|V|r\rangle|^2}{E_r - E_j}.$$

Hence second-order non-degenerate perturbation theory gives

$$\boxed{E = E_r + \lambda \langle r|V|r\rangle + \lambda^2 \sum_{j\neq r} \frac{|\langle j|V|r\rangle|^2}{E_r - E_j} + O(\lambda^3).}$$

One should be able to quote and/or derive this result.

Example: a quadratic perturbation of the harmonic oscillator

Consider

$$H = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 = \hbar\omega\left(a^\dagger a + \frac{1}{2}\right), \qquad E_n = \hbar\omega\left(n + \frac{1}{2}\right),$$

with states $|n\rangle$, $n = 0, 1, \ldots$, and perturbation $V = m\omega^2\hat{x}^2$. Using $\hat{x} = \sqrt{\hbar/(2m\omega)}(a + a^\dagger)$,

$$\begin{aligned} V &= m\omega^2\frac{\hbar}{2m\omega}\left(a^2 + (a^\dagger)^2 + aa^\dagger + a^\dagger a\right) \\ &= \frac{\hbar\omega}{2}\left(a^2 + (a^\dagger)^2 + 2a^\dagger a + 1\right). \end{aligned}$$

The nonzero matrix elements are

$$\begin{aligned} \langle n|V|n\rangle &= \frac{\hbar\omega}{2}(2n+1), \\ \langle n+2|V|n\rangle &= \frac{\hbar\omega}{2}\sqrt{(n+1)(n+2)}, \\ \langle n-2|V|n\rangle &= \frac{\hbar\omega}{2}\sqrt{n(n-1)} \qquad (n \geq 2). \end{aligned}$$

All other $\langle m|V|n\rangle$ vanish. For $n = 0, 1$, there is no $|n-2\rangle$ contribution.

The perturbed energy of the nth level through second order is

$$\begin{aligned} E_n(\lambda) &= E_n + \lambda\langle n|V|n\rangle + \lambda^2\sum_{m\neq n}\frac{|\langle m|V|n\rangle|^2}{E_n - E_m} + O(\lambda^3) \\ &= \hbar\omega\left(n + \frac{1}{2}\right) + \lambda\hbar\omega\left(n + \frac{1}{2}\right) \\ &\quad + \lambda^2\left(\frac{\hbar\omega}{2}\right)^2\left[\frac{(n+1)(n+2)}{-2\hbar\omega} + \frac{n(n-1)}{2\hbar\omega}\right] + O(\lambda^3) \\ &= \hbar\omega\left(n + \frac{1}{2}\right)\left(1 + \lambda - \frac{\lambda^2}{2}\right) + O(\lambda^3). \end{aligned}$$

This problem can also be solved exactly. Adding λV replaces the oscillator frequency by

$$\omega'^2 = \omega^2(1 + 2\lambda).$$

For $1 + 2\lambda > 0$, the exact energies are

$$\begin{aligned} E_n(\lambda) &= \hbar\omega'\left(n + \frac{1}{2}\right) = \hbar\omega\left(n + \frac{1}{2}\right)(1 + 2\lambda)^{1/2} \\ &= \hbar\omega\left(n + \frac{1}{2}\right)\left(1 + \lambda - \frac{1}{2}\lambda^2 + O(\lambda^3)\right), \end{aligned}$$

in agreement with the perturbative calculation.

End of lecture 11

Non-degenerate perturbation theory: recap

For the time-independent eigenvalue problem

$$(H + \lambda V)|\psi\rangle = E|\psi\rangle,$$

second-order non-degenerate perturbation theory gives

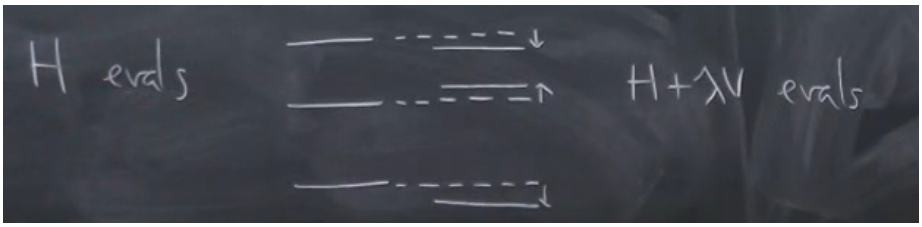
$$E = E_r + \lambda\langle r|V|r\rangle + \lambda^2 \sum_{j\neq r} \frac{|\langle j|V|r\rangle|^2}{E_r - E_j} + \cdots .$$

In general, we expect perturbation theory to be valid and useful when $|\lambda|$ is sufficiently small. But the corrections actually involve quantities such as

$$\lambda^2 \frac{|\langle j|V|r\rangle|^2}{|E_r - E_j|}$$

and similar expressions. The condition for accuracy is therefore that

$$|\text{energy shifts}| \ll |\text{original energy differences}|.$$



In general, the perturbation series may not converge, but can still be useful for approximate calculations as an asymptotic series. In the harmonic-oscillator example above, the total potential was

$$\frac{1}{2}m\omega^2(1 + 2\lambda)\hat{x}^{\,2}.$$

For $\lambda < -\frac{1}{2}$, this no longer describes a bound harmonic oscillator. This suggests that perturbation theory about $\lambda = 0$ converges, at best, for

$$|\lambda| < \frac{1}{2}.$$

Example: ground-state energy of helium

The unperturbed problem consists of two non-interacting electrons, each of charge $-e$, orbiting a nucleus of charge $+2e$. The Hamiltonian is

$$H_0 = H(\hat{\mathbf{x}}_1, \hat{\mathbf{p}}_1) + H(\hat{\mathbf{x}}_2, \hat{\mathbf{p}}_2),$$

where

$$H(\hat{\mathbf{x}}, \hat{\mathbf{p}}) = \frac{\hat{\mathbf{p}}^{\,2}}{2m} - \frac{2e^2}{4\pi\varepsilon_0|\hat{\mathbf{x}}|}.$$

The one-electron states are $|n\ell m\rangle$, with wavefunctions $\psi_{n\ell m}(\mathbf{x})$ and energies

$$E_n = -\frac{m}{2}\left(\frac{2e^2}{4\pi\varepsilon_0\hbar}\right)^2 \frac{1}{n^2}.$$

The lowest-energy state for the two electrons is

$$|\Psi\rangle = |100\rangle \otimes |100\rangle \otimes |\chi\rangle,$$

with the spin state

$$|\chi\rangle = \frac{1}{\sqrt{2}}\big(|\uparrow\rangle|\downarrow\rangle - |\downarrow\rangle|\uparrow\rangle\big).$$

The spatial part is symmetric and the spin part is antisymmetric, so the complete state is antisymmetric, as required for electrons. The spatial two-electron wavefunction is

$$\Psi(\mathbf{x}_1, \mathbf{x}_2) = \psi_{100}(\mathbf{x}_1)\psi_{100}(\mathbf{x}_2),$$

where the hydrogen-like solution with $Z = 2$ is

$$\psi_{100}(\mathbf{x}) = \frac{1}{\sqrt{\pi}}\left(\frac{1}{a_2}\right)^{3/2} e^{-|\mathbf{x}|/a_2}, \qquad a_2 = \frac{1}{2}\frac{4\pi\varepsilon_0\hbar^2}{me^2} = \frac{1}{2}\,(\text{Bohr radius}).$$

Thus the unperturbed ground-state energy is

$$2E_1 = -m\left(\frac{2e^2}{4\pi\varepsilon_0\hbar}\right)^2 \simeq -108.8\,\text{eV},$$

compared with $-13.6\,\text{eV}$ for hydrogen. Experimentally, the ground-state energy of helium is approximately $-79.0\,\text{eV}$, but we have neglected the electron–electron interaction.

Treat this interaction as a perturbation, writing

$$H_{\text{He}} = H_0 + \lambda V, \qquad \lambda = \frac{e^2}{4\pi\varepsilon_0}, \qquad V(\mathbf{x}_1, \mathbf{x}_2) = \frac{1}{|\mathbf{x}_1 - \mathbf{x}_2|}.$$

The first-order correction to the ground-state energy is

$$\begin{aligned}\lambda\langle\Psi|V|\Psi\rangle &= \lambda\int d^3x_1 \int d^3x_2\, \Psi(\mathbf{x}_1, \mathbf{x}_2)^* V(\mathbf{x}_1, \mathbf{x}_2)\Psi(\mathbf{x}_1, \mathbf{x}_2) \\ &= \lambda\int d^3x_1 \int d^3x_2\, \frac{|\psi_{100}(\mathbf{x}_1)|^2|\psi_{100}(\mathbf{x}_2)|^2}{|\mathbf{x}_1 - \mathbf{x}_2|} \\ &\simeq 34.0\,\text{eV}.\end{aligned}$$

Here the normalised spin factor contributes one, since V is independent of spin. The corrected ground-state energy is therefore

$$E_{\text{He}} \simeq -108.8\,\text{eV} + 34.0\,\text{eV} = -74.8\,\text{eV}.$$

4.2 Degenerate Case

Consider again

$$(H + \lambda V)|\psi\rangle = E|\psi\rangle,$$

and suppose that, as $\lambda \to 0$,

$$E \to E_r = E_s = \cdots, \qquad |\psi\rangle \to \sum_r a_r|r\rangle.$$

Use the labels $r, s, \ldots$ *only* for the degenerate states at this unperturbed energy. Use j *only* for the other basis states $|j\rangle$, whose energies satisfy $E_j \neq E_r$.

For general λ, expand

$$|\psi\rangle = \sum_r \alpha_r|r\rangle + \sum_j \beta_j|j\rangle.$$

We must have

$$\alpha_r = a_r + O(\lambda), \qquad \beta_j = O(\lambda).$$

Substituting into the eigenvalue equation gives

$$\sum_r \alpha_r(E_r + \lambda V)|r\rangle + \sum_j \beta_j(E_j + \lambda V)|j\rangle$$
$$= E\left(\sum_r \alpha_r|r\rangle + \sum_j \beta_j|j\rangle\right).$$

Rearranging,

$$\sum_r \alpha_r(E - E_r)|r\rangle + \sum_j \beta_j(E - E_j)|j\rangle$$
$$= \lambda\sum_r \alpha_r V|r\rangle + \lambda\sum_j \beta_j V|j\rangle. \qquad (**)$$

Seek

$$E = E_r + \lambda E' + O(\lambda^2).$$

The correction E' is no longer associated with a specific basis state in our chosen degenerate set.

Take the inner product of $(**)$ with $\langle s|$:

$$(E - E_s)\alpha_s = \lambda\sum_r \alpha_r\langle s|V|r\rangle + \lambda\sum_j \beta_j\langle s|V|j\rangle.$$

Since $E_s = E_r$, to leading order in λ we obtain

$$\boxed{\sum_r \langle s|V|r\rangle a_r = E' a_s.}$$

Thus E' is an eigenvalue of the matrix

$$\big(\langle s|V|r\rangle\big)$$

within the degenerate subspace, and the a_r are the components of the corresponding eigenvector. If there are N degenerate states, there are N solutions for E' (counting multiplicity) and corresponding eigenvectors. The zeroth-order states selected by the perturbation are

$$\sum_r a_r|r\rangle.$$

General picture

Degeneracy of H reflects some symmetry. The perturbed Hamiltonian $H + \lambda V$ generally has less symmetry, so the energy levels split apart.



Example: a particle in a two-dimensional box

Consider a particle in a square box,

$$0 \le x, y \le a.$$

The unperturbed states are $|pq\rangle$, with wavefunctions

$$\psi_{pq}(x,y) = \frac{2}{a} \sin\frac{p\pi x}{a} \sin\frac{q\pi y}{a}, \qquad p, q = 1, 2, \ldots.$$

The lowest energy level is

$$E_{11} = \frac{\hbar^2\pi^2}{ma^2}, \qquad \text{non-degenerate.}$$

The next level is

$$E_{12} = E_{21} = \frac{5\hbar^2\pi^2}{2ma^2}, \qquad \text{degeneracy 2.}$$

Take the perturbation

$$V(x,y) = \frac{xy}{a^2}, \qquad \lambda \text{ has dimensions of energy.}$$

For the lowest energy level,

$$\langle 11|V|11\rangle = \int_0^a dx \int_0^a dy\, \frac{xy}{a^2} |\psi_{11}(x,y)|^2 = \frac{1}{4}.$$

Hence the first-order energy shift is $\lambda/4$, and

$$E_{\text{ground}}(\lambda) = E_{11} + \frac{\lambda}{4} + O(\lambda^2).$$

For the degenerate level, diagonalise the matrix

$$\begin{pmatrix} \langle 12|V|12\rangle & \langle 12|V|21\rangle \\ \langle 21|V|12\rangle & \langle 21|V|21\rangle \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix}.$$

Its eigenvalues and normalised eigenvectors are

$$\alpha \pm \beta, \qquad \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}.$$

Thus the two first-order energy shifts are $\lambda(\alpha \pm \beta)$, with energies

$$E_\pm(\lambda) = E_{12} + \lambda(\alpha \pm \beta) + O(\lambda^2),$$

and the corresponding zeroth-order states are

$$|\psi_\pm^{(0)}\rangle = \frac{1}{\sqrt{2}}\big(|12\rangle \pm |21\rangle\big).$$

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The diagonal matrix element is

$$\begin{aligned}\alpha &= \int_0^a dx \int_0^a dy\, \frac{xy}{a^2} |\psi_{12}(x,y)|^2 \\ &= \left(\frac{2}{a}\right)^2 \frac{1}{a^2} \left(\int_0^a x \sin^2 \frac{\pi x}{a}\, dx\right) \left(\int_0^a y \sin^2 \frac{2\pi y}{a}\, dy\right) = \frac{1}{4},\end{aligned}$$

using

$$\int_0^a x \sin^2 \frac{n\pi x}{a}\, dx = \frac{a^2}{4}, \qquad n = 1, 2, \ldots.$$

The off-diagonal matrix element is

$$\begin{aligned}\beta &= \int_0^a dx \int_0^a dy\, \frac{xy}{a^2} \psi_{12}(x,y)\psi_{21}(x,y) \\ &= \left(\frac{2}{a}\right)^2 \frac{1}{a^2} \left(\int_0^a x \sin \frac{\pi x}{a} \sin \frac{2\pi x}{a}\, dx\right)^2 = \left(\frac{4}{3\pi}\right)^4,\end{aligned}$$

where

$$\int_0^a x \sin \frac{\pi x}{a} \sin \frac{2\pi x}{a}\, dx = -\frac{8a^2}{9\pi^2}.$$

Summing up, to first order the new ground-state energy is

$$E_{\text{ground}} = \frac{\hbar^2 \pi^2}{ma^2} + \frac{\lambda}{4} + O(\lambda^2).$$

The next two levels have energies

$$E_\pm = \frac{5\hbar^2\pi^2}{2ma^2} + \frac{\lambda}{4} \pm \lambda \left(\frac{4}{3\pi}\right)^4 + O(\lambda^2),$$

with corresponding zeroth-order states

$$\frac{1}{\sqrt{2}}\big(|12\rangle \pm |21\rangle\big).$$

We expect perturbation theory to be accurate for

$$|\lambda| \ll \frac{\hbar^2\pi^2}{ma^2}.$$

5. Angular Momentum

5.1 Recap of Orbital Angular Momentum

Consider a particle with position $\hat{\mathbf{x}}$ and momentum $\hat{\mathbf{p}}$. Its angular momentum about the origin is

$$\mathbf{L} = \hat{\mathbf{x}} \times \hat{\mathbf{p}}, \qquad L_i = \varepsilon_{ijk} \hat{x}_j \hat{p}_k.$$

Repeated Cartesian indices are summed. From

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij},$$

we find

$$[L_i, L_j] = i\hbar\varepsilon_{ijk} L_k.$$

The squared angular momentum

$$L^2 = L_i L_i$$

satisfies

$$[L^2, L_i] = 0.$$

The angular-momentum commutator can be derived by first showing

$$[L_i, \hat{x}_j] = i\hbar\varepsilon_{ijk}\hat{x}_k, \qquad [L_i, \hat{p}_j] = i\hbar\varepsilon_{ijk}\hat{p}_k.$$

The orbital angular-momentum eigenvalue problem has well-behaved solutions only for

$$\ell = 0, 1, 2, \ldots, \qquad m = 0, \pm 1, \pm 2, \ldots, \pm\ell.$$

If the Schrödinger equation has a spherically symmetric potential $V(\mathbf{x}) = V(r)$, seek separable solutions

$$\psi_{\ell m}(\mathbf{x}) = R_\ell(r) Y_{\ell m}(\theta, \varphi).$$

The time-independent Schrödinger equation for the radial function is

$$-\frac{\hbar^2}{2m}\frac{1}{r}\frac{d^2}{dr^2}\big(rR_\ell\big) + \left(\frac{\hbar^2\ell(\ell+1)}{2mr^2} + V(r)\right) R_\ell = ER_\ell.$$

Parity

Of particular importance is the behaviour under parity, $\mathbf{x} \mapsto -\mathbf{x}$. In spherical polar coordinates this is

$$r \mapsto r, \qquad \theta \mapsto \pi - \theta, \qquad \varphi \mapsto \varphi + \pi \pmod{2\pi}.$$

The spherical harmonics satisfy

$$Y_{\ell m}(\pi - \theta, \varphi + \pi) = (-1)^\ell Y_{\ell m}(\theta, \varphi).$$

Hence

$$\psi_{\ell m}(-\mathbf{x}) = (-1)^\ell \psi_{\ell m}(\mathbf{x}).$$

Two particles with a central interaction

An important case is a system of two particles interacting through a potential depending only on their relative separation:

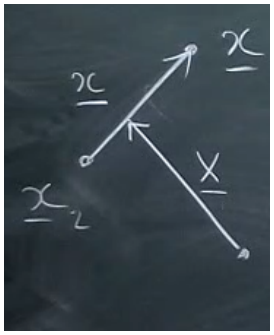
$$\begin{aligned} H &= \frac{\hat{\mathbf{p}}_1^{\,2}}{2m_1} + \frac{\hat{\mathbf{p}}_2^{\,2}}{2m_2} + V\big(|\hat{\mathbf{x}}_1 - \hat{\mathbf{x}}_2|\big) \\ &= \frac{\hat{\mathbf{P}}^{\,2}}{2M} + \frac{\hat{\mathbf{p}}^{\,2}}{2m} + V\big(|\hat{\mathbf{x}}|\big). \end{aligned}$$

The centre-of-mass variables are

$$\hat{\mathbf{X}} = \frac{m_1\hat{\mathbf{x}}_1 + m_2\hat{\mathbf{x}}_2}{M}, \qquad \hat{\mathbf{P}} = \hat{\mathbf{p}}_1 + \hat{\mathbf{p}}_2, \qquad M = m_1 + m_2.$$

The relative variables and reduced mass are

$$\hat{\mathbf{x}} = \hat{\mathbf{x}}_1 - \hat{\mathbf{x}}_2, \qquad \hat{\mathbf{p}} = \frac{m_2\hat{\mathbf{p}}_1 - m_1\hat{\mathbf{p}}_2}{M}, \qquad m = \frac{m_1 m_2}{M}.$$



The centre-of-mass dynamics are those of a free particle, since there is no external force or potential. The relative motion is governed by a spherically symmetric potential. A separated wavefunction is

$$\Psi(\mathbf{X}, \mathbf{x}) = e^{i\mathbf{k}\cdot\mathbf{X}}\psi_{\ell m}(\mathbf{x}).$$

Under $\mathbf{x} \mapsto -\mathbf{x}$ at fixed $\mathbf{X}$,

$$\Psi \mapsto (-1)^\ell \Psi.$$

Interchanging the particles, $\mathbf{x}_1 \leftrightarrow \mathbf{x}_2$, reverses $\mathbf{x} = \mathbf{x}_1 - \mathbf{x}_2$. For identical particles the centre of mass is unchanged, so the spatial wavefunction acquires the factor $(-1)^\ell$ under particle interchange.

5.2 General Analysis of Angular Momentum Eigenstates

Consider Hermitian operators

$$\mathbf{J} = (J_1, J_2, J_3), \qquad J_i^\dagger = J_i,$$

obeying

$$[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k.$$

Define

$$J^2 = J_iJ_i, \qquad (J^2)^\dagger = J^2.$$

Then

$$[J^2, J_i] = 0.$$

First observe that if $|\psi\rangle$ is normalised and

$$J^2|\psi\rangle = \lambda|\psi\rangle,$$

then

$$\begin{aligned}\lambda &= \langle\psi|J^2|\psi\rangle \\ &= \|J_1|\psi\rangle\|^2 + \|J_2|\psi\rangle\|^2 + \|J_3|\psi\rangle\|^2 \geq 0.\end{aligned}$$

It is convenient to write

$$\lambda = \hbar^2 j(j+1), \qquad j \geq 0,$$

since $j(j+1)$ is unchanged by $j \mapsto -(j+1)$. Label simultaneous eigenstates of J^2 and J_3 by $|jm\rangle$:

$$J^2|jm\rangle = \hbar^2 j(j+1)|jm\rangle, \qquad J_3|jm\rangle = \hbar m|jm\rangle.$$

To analyse the allowed values of j and m, define the ladder operators

$$J_\pm = J_1 \pm iJ_2, \qquad (J_\pm)^\dagger = J_\mp.$$

It is easy to check that

$$[J_3, J_\pm] = \pm\hbar J_\pm, \qquad [J_+, J_-] = 2\hbar J_3, \qquad [J^2, J_\pm] = 0.$$

In fact,

$$J_+J_- = J^2 - J_3^2 + \hbar J_3, \qquad J_-J_+ = J^2 - J_3^2 - \hbar J_3.$$

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Given a simultaneous eigenstate $|jm\rangle$, the state $J_\pm|jm\rangle$, provided it is non-zero, is again an eigenstate of J^2 with the same value of j, while its J_3 eigenvalue is shifted by $\pm\hbar$. Indeed,

$$\begin{aligned} J_3\big(J_\pm|jm\rangle\big) &= \big([J_3, J_\pm] + J_\pm J_3\big)|jm\rangle \\ &= \big(\pm\hbar J_\pm + J_\pm \hbar m\big)|jm\rangle \\ &= \hbar(m \pm 1) J_\pm|jm\rangle, \end{aligned}$$

and $[J^2, J_\pm] = 0$ gives

$$J^2\big(J_\pm|jm\rangle\big) = \hbar^2 j(j+1) J_\pm|jm\rangle.$$

Compare this with the harmonic oscillator, where

$$[N, a^\dagger] = a^\dagger, \qquad [N, a] = -a,$$

so $a^\dagger|n\rangle$ and $a|n\rangle$, when non-zero, have N eigenvalues $n+1$ and $n-1$, respectively.

Bounds on the magnetic quantum number

To determine when the new states vanish, take $|jm\rangle$ to be normalised and calculate their squared norms:

$$\begin{aligned} \|J_+|jm\rangle\|^2 &= \langle jm|J_- J_+|jm\rangle \\ &= \langle jm|J^2 - J_3^2 - \hbar J_3|jm\rangle \\ &= \hbar^2\big(j(j+1) - m(m+1)\big) \\ &= \hbar^2(j-m)(j+m+1) \geq 0. \end{aligned}$$

Similarly,

$$\begin{aligned} \|J_-|jm\rangle\|^2 &= \langle jm|J_+ J_-|jm\rangle \\ &= \langle jm|J^2 - J_3^2 + \hbar J_3|jm\rangle \\ &= \hbar^2\big(j(j+1) - m(m-1)\big) \\ &= \hbar^2(j+m)(j-m+1) \geq 0. \end{aligned}$$

Since $j \geq 0$, these inequalities imply, respectively,

$$-(j+1) \leq m \leq j, \qquad -j \leq m \leq j+1.$$

Combining them gives

$$\boxed{-j \leq m \leq j.}$$

Termination of the ladder

Starting from a normalised state $|jm\rangle$, each non-vanishing state $J_+^n|jm\rangle$ has J_3 eigenvalue $\hbar(m+n)$, subject to $m+n \leq j$. The raising sequence must therefore terminate. If $J_+^k|jm\rangle$ is the last non-zero state, its raised state has zero norm, so the norm formula and the bound on m give

$$j = m + k, \qquad k \in \mathbb{Z}_{\geq 0}.$$

Similarly, repeated lowering reaches a lowest state after some $k' \in \mathbb{Z}_{\geq 0}$ steps, with

$$-j = m - k'.$$

Consequently,

$$J_+^n|jm\rangle = 0 \quad \text{for all integers } n > k, \qquad J_-^n|jm\rangle = 0 \quad \text{for all integers } n > k'.$$

Thus

$$j = m + k = -m + k', \qquad 2j = k + k' \in \mathbb{Z}_{\geq 0}.$$

The allowed angular-momentum quantum numbers are therefore

$$j = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots,$$

and, for each fixed j, one angular-momentum multiplet consists of

$$|jm\rangle, \qquad m = j, j-1, \ldots, -j+1, -j,$$

giving $2j+1$ states. There are two possibilities:

- j integral: $m = 0, \pm 1, \ldots, \pm j$, with an odd number of states;
- j half-integral: $m = \pm\frac{1}{2}, \pm\frac{3}{2}, \ldots, \pm j$, with an even number of states.

These correspond to $2j = k + k'$ being even or odd.

The integral case is realised by orbital angular momentum. Its states are represented by wavefunctions $\psi_{jm}(\mathbf{x})$ with angular dependence $Y_{jm}(\theta, \varphi)$, as discussed in §5.1. Half-integral values do not arise from well-behaved scalar orbital wavefunctions. They occur for intrinsic angular momentum, or *spin*, which may have either integral or half-integral quantum numbers.

Normalised ladder states

We can choose the relative phases of the normalised states so that

$$J_+|jm\rangle = \hbar\sqrt{(j-m)(j+m+1)}\,|j, m+1\rangle,$$
$$J_-|jm\rangle = \hbar\sqrt{(j+m)(j-m+1)}\,|j, m-1\rangle.$$

All states in a multiplet can be generated from any one of them using the ladder operators. It is often convenient to start with the *top state*, which has the largest J_3 eigenvalue and satisfies

$$J_+|jj\rangle = 0.$$

All other states are obtained, up to normalisation, as

$$(J_-)^n|jj\rangle, \qquad n = 0, 1, \ldots, 2j.$$

5.3 Matrix Representations

We now express the results of §5.2 using column vectors for states and matrices for operators. Recall that, in an orthonormal basis,

$$|\psi\rangle \longleftrightarrow (\alpha_n), \qquad \alpha_n = \langle n|\psi\rangle, \qquad A_{mn} = \langle m|A|n\rangle.$$

$j = 1$

The state space of a single $j = 1$ multiplet is three-dimensional. Choose the ordered basis

$$|1,1\rangle \longleftrightarrow \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad |1,0\rangle \longleftrightarrow \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad |1,-1\rangle \longleftrightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Then

$$J_3 = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad J_+ = \hbar\sqrt{2} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad J_- = \hbar\sqrt{2} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

These follow from

$$J_3|1,1\rangle = \hbar|1,1\rangle, \qquad J_3|1,0\rangle = 0, \qquad J_3|1,-1\rangle = -\hbar|1,-1\rangle,$$

and the ladder relations above. In particular,

$$J_\pm|1,0\rangle = \hbar\sqrt{2}\,|1,\pm1\rangle.$$

Also,

$$J_\mp|1,\pm1\rangle = \hbar\sqrt{2}\,|1,0\rangle.$$

$j = s = \frac{1}{2}$

The spin-$\frac{1}{2}$ state space is two-dimensional. Choose

$$\left|\frac{1}{2},\frac{1}{2}\right\rangle = |\uparrow\rangle \longleftrightarrow \begin{pmatrix}1\\0\end{pmatrix}, \qquad \left|\frac{1}{2},-\frac{1}{2}\right\rangle = |\downarrow\rangle \longleftrightarrow \begin{pmatrix}0\\1\end{pmatrix}.$$

The spin operators are represented by

$$S_3 = \frac{\hbar}{2}\begin{pmatrix}1&0\\0&-1\end{pmatrix}, \qquad S_+ = \hbar\begin{pmatrix}0&1\\0&0\end{pmatrix}, \qquad S_- = \hbar\begin{pmatrix}0&0\\1&0\end{pmatrix}.$$

Equivalently,

$$S_i = \frac{\hbar}{2}\sigma_i,$$

where the Pauli matrices are

$$\sigma_1 = \begin{pmatrix}0&1\\1&0\end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix}0&-i\\i&0\end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix}1&0\\0&-1\end{pmatrix}.$$

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Recall that the angular-momentum states $|jm\rangle$ form a multiplet of $2j+1$ states for each integral or half-integral j. For spin $\frac{1}{2}$, the Pauli matrices above are Hermitian, traceless 2×2 matrices satisfying

$$\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \mathbf{1}, \qquad \sigma_1\sigma_2 = -\sigma_2\sigma_1 = i\sigma_3$$

and cyclic permutations of the second relation. In index notation,

$$\sigma_i\sigma_j = \delta_{ij}\mathbf{1} + i\epsilon_{ijk}\sigma_k.$$

The commutator $[\sigma_i,\sigma_j] = 2i\epsilon_{ijk}\sigma_k$ gives $[S_i,S_j] = i\hbar\epsilon_{ijk}S_k$. The symmetric product $\sigma_i\sigma_j + \sigma_j\sigma_i = 2\delta_{ij}\mathbf{1}$ is special to the spin-$\frac{1}{2}$ representation.

Writing $\boldsymbol{\sigma} = (\sigma_1,\sigma_2,\sigma_3)$ and $\mathbf{S} = \frac{\hbar}{2}\boldsymbol{\sigma}$, we have

$$(\mathbf{a}\cdot\boldsymbol{\sigma})(\mathbf{b}\cdot\boldsymbol{\sigma}) = (\mathbf{a}\cdot\mathbf{b})\mathbf{1} + i(\mathbf{a}\times\mathbf{b})\cdot\boldsymbol{\sigma}$$

for constant vectors $\mathbf{a},\mathbf{b}$, or operator-valued vectors whose components commute with each S_i. For any real unit vector $\mathbf{n}$ this yields

$$(\mathbf{n}\cdot\boldsymbol{\sigma})^2 = \mathbf{1}, \qquad (\mathbf{n}\cdot\mathbf{S})^2 = \frac{\hbar^2}{4}\mathbf{1}.$$

Since $\mathbf{n}\cdot\mathbf{S}$ is traceless, its two eigenvalues are $\pm\frac{\hbar}{2}$ for every direction $\mathbf{n}$. Also,

$$\boldsymbol{\sigma}^2 = \sigma_1^2+\sigma_2^2+\sigma_3^2 = 3\mathbf{1}, \qquad \mathbf{S}^2 = \frac{3\hbar^2}{4}\mathbf{1} = \hbar^2\frac{1}{2}\left(\frac{1}{2}+1\right)\mathbf{1}.$$

5.4 Implications of Angular Momentum and Spin

(a) Orbital, spin and total angular momentum

The internal states $|r\rangle$ introduced earlier are spin states $|s, r\rangle$ with $r = -s, -s+1, \ldots, s$. The spin operators satisfy

$$[S_i, \hat{x}_j] = [S_i, \hat{p}_j] = 0,$$

consistent with the product basis $|\mathbf{x}\rangle \otimes |r\rangle$ for a particle of spin s. The operators $\hat{\mathbf{x}}$, $\mathbf{S}^2$ and S_3 form a complete commuting set. Orbital angular momentum $\mathbf{L}$ has nontrivial commutators with position and momentum.

The total angular momentum is $\mathbf{J} = \mathbf{L} + \mathbf{S}$. Since $[S_i, L_j] = 0$, the separate relations

$$[S_i, S_j] = i\hbar\epsilon_{ijk}S_k, \qquad [L_i, L_j] = i\hbar\epsilon_{ijk}L_k$$

give $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$.

(b) Interactions and observations

A magnetic dipole moment $\boldsymbol{\mu}$ in a magnetic field $\mathbf{B}$ has interaction energy $-\boldsymbol{\mu}\cdot\mathbf{B}$. The energy is lowest when the dipole is aligned with the field and highest when it is anti-aligned. A particle of charge q and mass m in an orbit with angular momentum $\mathbf{L}$ has

$$\boldsymbol{\mu} = \frac{q}{2m}\mathbf{L}, \qquad H_{\text{int}} = -\gamma\,\mathbf{L}\cdot\mathbf{B}, \qquad \gamma = \frac{q}{2m}.$$

For hydrogen, or a particle in another spherically symmetric potential, choose simultaneous eigenstates of the zero-field Hamiltonian H, $\mathbf{L}^2$ and L_3. At fixed radial quantum number and ℓ, their energy E is independent of m. A weak magnetic field along the 3-direction gives, to first order in B,

$$H \longmapsto H - \gamma B L_3.$$

The same states are eigenstates of this Hamiltonian, with energies

$$E - \gamma B\hbar m, \qquad m = -\ell, -\ell+1, \ldots, \ell.$$

The splitting of the orbital multiplet into $2\ell+1$ levels is the *Zeeman effect*.

The corresponding spin interaction is

$$H_{\text{spin}} = -\gamma'\mathbf{S}\cdot\mathbf{B},$$

where γ' can differ from γ. For a spin-$\frac{1}{2}$ particle with degenerate spin states at zero field, the shifts for $S_3 = \pm\hbar/2$ are $\mp\gamma' B\hbar/2$. This Zeeman doublet provides experimental evidence for spin $\frac{1}{2}$. If only spin degrees of freedom are relevant, H_{spin} can be used as the Hamiltonian in the background field.

(i) In the Stern–Gerlach experiment, an inhomogeneous magnetic field produces different forces on different spin projections, separating the beams in space.

(ii) The orbital and spin magnetic moments of the electron lead to a spin–orbit term proportional to $\mathbf{L}\cdot\mathbf{S}$ in the Hamiltonian. It is of the same order as the first relativistic correction to the kinetic energy. Expanding in momentum gives

$$\sqrt{m^2c^4 + p^2c^2} - mc^2 = \frac{p^2}{2m} - \frac{p^4}{8m^3c^2} + O\!\left(\frac{p^6}{m^5c^4}\right).$$

These effects contribute to the small energy splittings called *fine structure*.

(iii) A splitting also occurs in hydrogen with no external magnetic field and with $L = 0$.

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